

3

Markets for Electrical Energy

3.1 Introduction

As a first step in our study of electricity markets, we will assume that all generators and loads are connected to the same bus or that they are connected through a lossless network of infinite capacity. We will, therefore, ignore for now the complexities introduced by the transmission and distribution networks and concentrate on the trading of electrical energy.

Since it is currently not economical to store large quantities of electrical energy, this energy must be produced at pretty much the same time as it is consumed. Trade in electrical energy, therefore, always refers to a certain amount of megawatt-hours to be delivered *over a specified period of time*. The length of this period of time is typically set at an hour, half an hour or quarter of an hour depending on the country or region where the market is located. Since electrical energy delivered during one period is not the same commodity as electrical energy delivered during another period, the price will usually be different for each period. Demand, however, does not change neatly at the beginning of each period. Some adjustments in production must therefore be made on a much shorter basis to keep the system in balance. While such adjustments translate into trades of energy, they are best treated as services rather than commodities, and we will postpone their detailed analysis until Chapter 5.

3.2 What is the Difference Between a Megawatt-hour and a Barrel of Oil?

The development of electricity markets is based on the premise that electrical energy can be treated as a commodity. There are, however, important differences between electrical energy and other commodities such as bushels of wheat, barrels of oil or even cubic meters of gas. These differences have a profound effect on the organization and the rules of electricity markets.

The most fundamental difference is that electrical energy is inextricably linked with a physical system that functions much faster than any market. In this physical

power system, supply and demand – generation and load – must be balanced on a second-by-second basis. If this balance is not maintained, the system collapses with catastrophic consequences. Such a breakdown is intolerable because it is not only the trading system that stops working but also an entire region or country that may be without power for many hours. Restoring a power system to normal operation following a complete collapse is a very complex process that may take 24 h or more in large, industrialized countries. The social and economic consequences of such a systemwide blackout are so severe that no sensible government would agree to the implementation of a market mechanism that significantly increases the likelihood of such an event. Balancing the supply and the demand for electrical energy in the short run is thus a process that simply cannot be left to a relatively slow-moving and unaccountable entity such as a market. In the short run, this balance must be maintained, at practically any cost, through a mechanism that does not rely on a market to select and dispatch resources.

Another significant (but somewhat less fundamental) difference between electrical energy and other commodities is that the energy produced by one generator cannot be directed to a specific consumer. Conversely, a consumer cannot take energy from only one generator. Instead, the power produced by all generators is pooled on its way to the loads. This pooling is possible because units of electrical energy produced by different generators are indistinguishable. Pooling is desirable because it results in valuable economies of scale: the maximum generation capacity must be commensurate with the maximum aggregated demand rather than with the sum of the maximum individual demands. On the other hand, a breakdown in a system in which the commodity is pooled affects everybody, not just the parties to a particular transaction.

Finally, the demand for electrical energy exhibits predictable daily and weekly cyclical variations. However, it is by no means the only commodity for which the demand is cyclical. The consumption of coffee, to take a simple example, exhibits two or three rather sharp peaks every day, separated by periods of lower demand. Trading in coffee does not require special mechanisms because consumers can easily store it in solid or liquid form. On the other hand, electrical energy must be produced at the same time as it is consumed. Since its short-run price elasticity of demand is extremely small, matching supply and demand requires production facilities capable of following the large and rapid changes in consumption that take place over the course of a day. Not all of these generating units will be producing throughout the day. When the demand is low, only the most efficient units are likely to be competitive and the others will be shut down temporarily. These less efficient units are needed only to supply the peak demand. Since the marginal producer changes as the load increases and decreases, we should expect the marginal cost of producing electrical energy (and hence the spot price of this energy) to vary over the course of the day. Such rapid cyclical variations in the cost and price of a commodity are very unusual.

One could argue that trading in gas also takes place over a physical network in which the commodity is pooled and the demand is cyclical. However, the amount of energy stored in the gas pipelines is considerably larger than the amount of kinetic energy stored in electricity-generating units. An imbalance between production and consumption of gas would therefore have to last much longer before it would cause a collapse of the pipeline network. Unlike an imbalance in a power system, it can be corrected through a market mechanism.

3.3 The Need for a Managed Spot Market

As discussed in Chapter 2, a market is an environment designed to help buyers and sellers interact and agree on transactions. These interactions progressively lead to an equilibrium in which the price clears the market, that is, the supply is equal to the demand. If electrical energy is to be traded according to this free-market ideal, the equilibrium between the production and the consumption of electrical energy should be set through the direct interaction of buyers and sellers.

In this ideal market, large consumers and retailers purchase electrical energy from generating companies. Like all rational consumers, they have to estimate how much to purchase. To this end, they forecast their consumption or the consumption of their own customers for every market period (hour, half hour or quarter of an hour) before entering into contracts. For their part, generators schedule the production of their units to deliver at the agreed time the energy that they have sold. Each generator clearly tries to minimize the cost of producing this energy. In practice, however, things are not that simple. Neither party can reliably meet its contractual obligations with perfect accuracy. First, the actual demand of a group of consumers is never exactly equal to the value forecasted. Second, unpredictable problems often prevent generating units from delivering the contracted amount of energy. A sudden mechanical or electrical failure may force a unit to shut down or reduce its output. More mundane problems can delay the synchronization of a unit to the system and hence affect the timing of its production of energy.

These errors and unpredictable events introduce gaps between load and generation that must be bridged quickly and precisely to maintain the integrity of the power system. If these gaps between generation and load were to be treated as imbalances between supply and demand and corrected using an open market mechanism, producers and consumers would have to be kept informed of the state of the market (offer, demand, prices) on a second-by-second basis. A sufficiently large number of them would have to be willing to trade on this timescale. They would also have to be able to adjust their production or consumption at any time and at short notice to absorb any credible imbalance. In the current state of the technology, it is difficult to conceive a system capable of transmitting the vast amounts of data required and of recording the thousands of transactions involved. Even if such an information infrastructure could be put in place, it remains to be proven that such a system would be sufficiently fast and reliable to prevent imbalances that might lead to a collapse of the entire power system. Finally, the transaction costs associated with such a system would be prohibitive.

We can therefore conclude that, while a large proportion of the electrical energy can be traded through an unmanaged open market, such a market is unable to maintain the reliability of the power system. A managed spot market that provides a mechanism for balancing load and generation must therefore supersede the open energy market as the time of delivery approaches. Its function is to match residual load and generation by adjusting the production of flexible generators and curtailing the demand of willing consumers. It should also be able to respond to major disruptions caused by the sudden and unforeseen disconnection of large generating units because of unavoidable technical problems. Although the need for managing the spot market stems from technical considerations, this spot market must operate in an economically efficient manner. Being out of balance may be unavoidable for producers and consumers, but

it should not be cost-free. To encourage efficient behavior, producers and consumers must pay the true cost of the electrical energy that is bought or sold in the spot market to restore the balance between load and generation.

Once a fair and efficient spot market is in place, electrical energy can be traded like other commodities. In the next section, we will discuss how this trade can be organized. We will then examine in more detail the design of a managed spot market and its interactions with the open electrical energy market.

3.4 Open Electrical Energy Markets

3.4.1 Bilateral trading

As its name implies, bilateral trading involves only two parties: a buyer and a seller. Participants thus enter into contracts without involvement, interference or facilitation from a third party. Depending on the amount of time available and the quantities to be traded, buyers and sellers will resort to different forms of bilateral trading:

Customized long-term contracts The terms of such contracts are flexible since they are negotiated privately to meet the needs and objectives of both parties. They usually involve the sale of large amounts of power (hundreds or thousands of MW) over long periods of time (several months to several years). The large transaction costs associated with the negotiation of such contracts make them worthwhile only when the parties want to buy or sell large amounts of energy.

Trading “over the counter” These transactions involve smaller amounts of energy to be delivered according to a standard profile, that is, a standardized definition of how much energy should be delivered during different periods of the day and week. This form of trading has much lower transaction costs and is used by producers and consumers to refine their position as delivery time approaches.

Electronic trading Participants can enter offers to buy energy and bids to sell energy directly in a computerized marketplace. All market participants can observe the quantities and prices submitted but do not know the identity of the party that submitted each bid or offer. When a party enters a new bid, the software that runs the exchange checks to see if there is a matching offer for the period of delivery of the bid. If it finds an offer whose price is greater than or equal to the price of the bid, a deal is automatically struck and the price and quantity are displayed for all participants to see. If no match is found, the new bid is added to the list of outstanding bids and will remain there until a matching offer is made or the bid is withdrawn or it lapses because the market closes for that period. A similar procedure is used each time a new offer is entered in the system. This form of trading is extremely fast and cheap. A flurry of trading activity often takes place in the minutes and seconds before the closing of the market as generators and retailers fine-tune their position ahead of the delivery period.

The essential characteristic of these three forms of bilateral trading is that the price of each transaction is set independently by the parties involved. There is thus no “official” price. While the details of negotiated long-term contracts are usually kept private, some independent reporting services usually gather information about over-the-counter trading and publish summary information about prices and quantities in

a form that does not reveal the identity of the parties involved. This type of market reporting and the display of the last transaction arranged through electronic trading enhance the efficiency of the market by giving all participants a clearer idea of the state and the direction of the market.

3.4.1.1 Example 3.1

Borduria Power trades in the Bordurian electricity market that operates on a bilateral basis. It owns the three generating units whose characteristics are given in the table below. To keep things simple, we have assumed that the marginal cost of these units is constant over their range of operation. Because of their large start-up cost, Borduria Power tries to keep unit A synchronized to the system at all times and to produce as much as possible with unit B during the daytime. The start-up cost of unit C is assumed to be negligible.

Unit	Type	P _{min}	P _{max}	MC
		(MW)	(MW)	(\$/MWh)
A	Large coal	100	500	10.0
B	Medium coal	50	200	13.0
C	Gas turbine	0	50	17.0

Let us focus on the contractual position of Borduria Power for the period between 2:00 and 3:00 P.M. on 11 June. The table below summarizes the relevant bilateral contracts.

Type	Contract date	Identifier	Buyer	Seller	Amount	Price
					(MWh)	(\$/MWh)
Long term	10 January	LT1	Cheapo Energy	Borduria Power	200	12.5
Long term	7 February	LT2	Borduria Steel	Borduria Power	250	12.8
Future	3 March	FT1	Quality Electrons	Borduria Power	100	14.0
Future	7 April	FT2	Borduria Power	Perfect Power	30	13.5
Future	10 May	FT3	Cheapo Energy	Borduria Power	50	13.8

Note that Borduria Power has taken advantage of the price fluctuations in the forward market to buy back at a profit some of the energy that it had sold. Toward midmorning on 11 June, Fiona, the trader on duty at Borduria Power, must decide if she wants to adjust this position by trading on the screen-based Bordurian Power Exchange (BPeX).

On the one hand, Borduria Power has contracted to deliver 570 MWh and has a total production capacity of 750 MW available during that hour. On the other hand, her BPeX trading screen displays the following stacks of bids and offers:

11 June 14:00 to 15:00	Identifier	Amount	Price
		(MW)	(\$/MWh)
Bids to sell energy	B5	20	17.50
	B4	25	16.30
	B3	20	14.40
	B2	10	13.90
	B1	25	13.70
Offers to buy energy	O1	20	13.50
	O2	30	13.30
	O3	10	13.25
	O4	30	12.80
	O5	50	12.55

On the basis of her experience with this market, Fiona believes that it is unlikely that the offer prices will increase. Since she still has 130 MW of spare capacity on unit B, she decides to grab offers O1, O2 and O3 before one of her competitors does. These offers are indeed profitable because their price is higher than the marginal cost of unit B. After completing these transactions, Fiona sends revised production instructions for this hour to the power plants. Unit A is to generate at rated power (500 MW), while unit B is to set its output at 130 MW and unit C is to remain on standby.

Shortly before the BPeX closes trading for the period between 14:00 and 15:00, Fiona receives a phone call from the operator of plant B. He informs her that the plant has developed some unexpected mechanical problems. It will be able to remain on-line until the evening but will not be able to produce more than 80 MW. Fiona quickly realizes that this failure leaves Borduria Power exposed and that she has three options:

- (1) Do nothing, leaving Borduria Power short by 50 MWh that would have to be paid for at the spot market price
- (2) Make up this deficit by starting up unit C
- (3) Try to buy some replacement power on the BPeX.

Since the spot market prices have been rather erratic lately, Fiona is not very keen on remaining unbalanced. She therefore decides to see if she can buy energy on the

BPeX for less than the marginal cost of unit C. Since she last traded on the BPeX, some bids have disappeared and new ones have been entered

11 June 14:00 to 15:00	Identifier	Amount	Price
		(MW)	(\$/MWh)
Bids to sell energy	B5	20	17.50
	B4	25	16.30
	B3	20	14.40
	B6	20	14.30
	B8	10	14.10
Offers to buy energy	O4	30	12.80
	O6	25	12.70
	O5	50	12.55

Fiona immediately selects bids B8, B6 and B3 because they allow her to restore the contractual balance of the company for this trading period at a cost that is less than the cost of covering the deficit with unit C. On balance, when trading closes for this hour, Borduria Power is committed to producing 580 MWh. Note that Fiona based all her decision on the incremental cost of producing energy. We will revisit this example below, once we have discussed the operation of the spot market.

3.4.2 Electricity pools

In the early days of the introduction of competition in electrical energy trading, bilateral trading was seen as too big a departure from the existing practice. Since electrical energy is pooled as it flows from the generators to the loads, it was felt that trading might as well be done in a centralized manner and involve all producers and consumers. Competitive electricity pools were thus created. Pools are a very unusual form of commodity trading but they have well-established roots in the operation of large power systems. In fact, some of the competitive electricity pools currently in operation were developed on the basis of collaborative pools created by monopoly utility companies with adjacent service territories.

Rather than relying on repeated interactions between suppliers and consumers to reach the market equilibrium, a pool provides a mechanism for determining this equilibrium in a systematic way. While there are many possible variations, a pool essentially operates as follows:

- Generating companies submit bids to supply a certain amount of electrical energy at a certain price for the period under consideration. These bids are ranked in order

of increasing price. From this ranking, a curve showing the bid price as a function of the cumulative bid quantity can be built. This curve is deemed to be the supply curve of the market.

- Similarly, the demand curve of the market can be established by asking consumers to submit offers specifying quantity and price and ranking these offers in decreasing order of price. Since the demand for electricity is highly inelastic, this step is sometimes omitted and the demand is set at a value determined using a forecast of the load. In other words, the demand curve is assumed to be a vertical line at the value of the load forecast.
- The intersection of these “constructed” supply and demand curves represents the market equilibrium. All the bids submitted at a price lower than or equal to the market clearing price are accepted and generators are instructed to produce the amount of energy corresponding to their accepted bids. Similarly, all the offers submitted at a price greater than or equal to the market clearing price are accepted and the consumers are informed of the amount of energy that they are allowed to draw from the system.
- The market clearing price represents the price of one additional megawatt-hour of energy and is therefore called the system marginal price or SMP. Generators are paid this SMP for every megawatt-hour that they produce, whereas consumers pay the SMP for every megawatt-hour that they consume, irrespective of the bids and offers that they submitted.

Paying the SMP for all the generation that was accepted may appear surprising at first glance. Why shouldn't generators that were willing to produce for less be paid only their asking price? Wouldn't this approach reduce the average price of electricity? The main reason this pay-as-bid scheme is not adopted is that it would discourage generators from submitting bids that reflect their marginal cost of production. All generators would instead try to guess what the SMP is likely to be and would bid at that level to collect the maximum revenues. At best, the SMP would therefore remain unchanged. Inevitably, some low-cost generators would occasionally overestimate the value of SMP and bid too high. These generators would then be left out of the schedule and be replaced by generators with a higher marginal cost of production. The SMP would then be somewhat higher than it ought to be. Furthermore, this substitution is economically inefficient because optimal use is not made of the available resources. In addition, generators are likely to increase their prices slightly to compensate themselves for the additional risk of losing revenue because of the uncertainty on the SMP. An attempt to reduce the price of electricity would therefore result in a price increase!

3.4.2.1 Example 3.2

The electricity pool of Syldavia has received the bids and offers shown on the table below for the period between 9:00 and 10:00 A.M. on 11 June.

Bids	Company	Quantity	Price
		(MWh)	(\$/MWh)
	Red	200	12.00
	Red	50	15.00
	Red	50	20.00
	Green	150	16.00
	Green	50	17.00
	Blue	100	13.00
	Blue	50	18.00
Offers	Yellow	50	13.00
	Yellow	100	23.00
	Purple	50	11.00
	Purple	150	22.00
	Orange	50	10.0
	Orange	200	25.00

Figure 3.1 shows how these bids and offers stack up to form the supply and demand curves respectively. From the intersection of these two curves, we conclude that for this particular period the SMP will be set at 16.00 \$/MWh and that 450 MWh will be traded through the Syldavian electricity pool. The table below shows how much energy each generator will be instructed to produce and how much energy each consumer will be allowed to draw. It also shows the revenues and expenses for each company

Company	Production	Consumption	Revenue	Expense
	(MWh)	(MWh)	(\$)	(\$)
Red	250		4000	
Blue	100		1600	
Green	100		1600	
Orange		200		3200
Yellow		100		1600
Purple		150		2400
Total	450	450	7200	7200

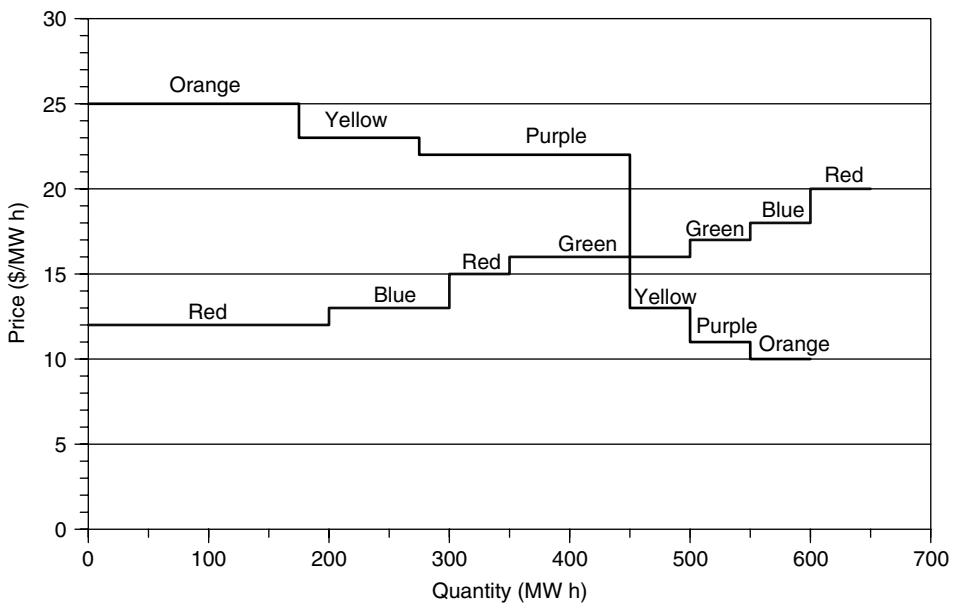


Figure 3.1 Stacks of bids and offers of Example 3.2

If, instead of asking consumers to submit offers, the Syldavian pool relied on a load forecast to represent the demand side and if the load for this period was forecasted to be 450 MWh, we would obtain exactly the same results.

In this example, generators submit simple bids consisting of price/quantity pairs. In some pools, generators submit complex bids for each of their generating units. These bids are supposed to reflect the cost characteristics of the unit (including the marginal, start-up and no-load costs) as well as some technical parameters (minimum and maximum output, flexibility). Rather than simply stacking the bids, the pool then performs a unit commitment calculation that determines the production schedule and the prices for an entire day divided in periods of half an hour or an hour. This approach to scheduling and pricing was used between 1990 and 2001 in the Electricity Pool of England and Wales.

3.4.3 Comparison of pool and bilateral trading

Since both the pool and the bilateral models of electrical energy trading have been adopted for electricity markets, it is worth reviewing the perceived advantages and disadvantages of both approaches.

As mentioned above, a competitive electricity pool is often created on the basis of an existing cooperation agreement between various utilities. Its conversion to operation on a competitive basis will therefore be less of a revolution than the creation of a completely new structure. Some of the concerns that accompany the introduction of competition may be alleviated by the somewhat less radical nature of the change. In particular, the public and the government are likely to have fewer concerns about the security of the electricity supply if the same organization remains in charge. A

pool provides a much more centralized form of system management. Not only does it handle all the physical electrical energy transactions but it usually also assumes the responsibility for operating the transmission system. This combination of roles avoids the multiplication of organizations but makes it more difficult to distinguish between the various functions that need to be performed in an electricity market.

Most small and medium electricity consumers have very little incentive to take an active part in an electricity market. Even when they are aggregated, the retailer that represents them has no direct means of adjusting consumption in response to changes in prices. One might therefore argue that the transaction costs could be reduced significantly if the demand is deemed to be passive and is represented by a load forecast in an electricity pool. Many economists are unhappy with this approach because they feel that direct negotiations between consumers and producers are essential if efficient prices are to be reached. Some economists dislike pools simply because they are only administered approximations of a market and not true markets.

Pools also provide a mechanism for reducing the scheduling risk faced by generators and hence, hopefully, the cost of electrical energy. When a generator sells energy on the basis of simple bids, for each market period separately, it runs the risk that for some periods it may not have sold enough energy to keep the plant on-line. At that point, it must decide whether to sell energy at a loss to keep the unit running or to shut it down and face the expense of another start-up at a later time. Either option increases the cost of producing energy with this unit and forces the generator to raise its average bid price. If this generator trades in a pool that operates on the basis of complex bids, the rules of this pool probably ensure that it recovers the start-up and no-load components of its bid. Moreover, the scheduling algorithm implemented by the pool usually tries to avoid unnecessary shutdowns. Since these factors reduce the risks faced by the generators, one would expect that they should foster lower average prices. This reduction in risk, however, comes at the price of an increase in the complexity of the pool rules. More complex rules reduce the transparency of the price setting process and increase opportunities for price manipulations. In practice, it is not clear whether complex bids and pool-based scheduling actually lower electricity prices.

3.5 The Managed Spot Market

For every commodity, imbalances almost always arise between the amount that a party has contracted to buy or sell and the amount that it actually needs or can produce. Spot markets provide a mechanism for handling these imbalances. If electrical energy is to be treated as a commodity, a spot market must therefore be organized. Unfortunately, as we discussed above, imbalances between generation and load must be corrected so quickly that a conventional spot market mechanism is not feasible. Instead, the system operator (SO) is given the responsibility to maintain the system in balance using what one might call a “managed spot market”. This mechanism is a market because the energy that is used to achieve this balance is freely offered by the participants at a price of their own choosing. It is a spot market because it determines the price at which imbalances are settled. However, it is a managed market because the bids and offers are selected by a third party (the SO) rather than through bilateral deals.

In the next paragraphs, we discuss the functionality of a generic managed spot market for electricity. Actual implementations can differ substantially from this blueprint.

There is also no consensus on the name to be given to this function. In addition to the term “spot market”, names such as “reserve market”, “balancing mechanism” and others are used.

3.5.1 Obtaining balancing resources

If market participants were able to predict with enough lead time and with perfect accuracy the amount of energy that they will consume or produce, the SO would not have to take balancing actions. The participants themselves could trade to cover their deficits and absorb their surpluses. In practice, there are always small imbalances and the SO must obtain adjustments in generation or load. Integrated over time, these adjustments translate into purchases and sales of electrical energy that can be settled at a spot price reflecting the market’s willingness to provide these adjustments. In keeping with the free market philosophy, any party that is willing to adjust its production or consumption should be allowed to do so on a competitive basis. This should give the SO the widest possible choice of balancing options and should therefore help reduce the cost of balancing. These balancing resources can be offered either for a specific period or on a long-term basis. Balancing services for a specific period are normally offered by market participants to the SO after the open energy market for that period has closed. Generating units that are not fully loaded can submit bids to increase their output. A generating unit can also offer to pay to reduce its output. This is a profitable proposition if the incremental price of this offer is less than the incremental cost of producing energy with this unit. A generating unit that submits such an offer is, in effect, trying to replace its own generation by cheaper power purchased on the spot market.

The demand side can also provide balancing resources. A consumer would offer to reduce its consumption if the price is greater than the value it places on consuming electricity during that period. Such demand reductions have the advantage that they can be implemented very quickly. It is also conceivable that consumers might offer to increase their demand if the price is sufficiently low.

Since these offers of balancing resources are submitted shortly before real time, the SO may be concerned about the amount or the price of balancing resources that will be offered. To guard against such problems, the SO can purchase balancing resources on a long-term basis. Under such contracts, the supplier is paid a fixed price (often called the option fee) to keep available some generation capacity. The contract also specifies the price or exercise fee to be paid for each MWh produced at the request of the SO using this capacity. The SO would call upon this contract only if the exercise fee is lower than what it would have to pay for a similar balancing resource offered on a short-term basis. As the terminology suggests, these contracts are equivalent to the option contracts used in financial and commodity markets. Their purpose is the same: to protect the buyer (in this case the SO) against price increases while guaranteeing some revenues to the supplier.

Imbalances due to forecasting errors by the participants are relatively small, evolve gradually and can be predicted to a certain extent. On the other hand, the imbalances that are caused by failures are often large, unpredictable and sudden. Many generating units can adjust their output at a rate that is sufficient to cope with the first type of imbalances. Handling the second type of imbalances requires generating units that can

increase their output rapidly and sustain this increased output for a certain time. We will explore the issue of reserve generation capacity in more detail when we discuss system security in Chapter 5. In the meantime, it is important to realize that all the units of energy that are traded to keep the system in balance do not have the same value. A megawatt obtained by slightly increasing the output of a large thermal plant costs considerably less than a megawatt of load that must be shed to prevent the system from collapsing. To be able to keep the system in balance at minimum cost, the SO should therefore have access to a variety of balancing resources. When producers and consumers bid to supply balancing resources, their bids must specify not only a quantity and a price but also how quickly the change in power injection can be delivered.

3.5.2 Gate closure

As we argued above, energy trading must stop at some point before real time to give the SO time to balance the system. How much time should elapse between this gate closure and real time is a hotly debated issue. System operators prefer longer intervals because this gives them more time to develop their plans and more flexibility in their selection of balancing resources. For example, if the gate closes half an hour before real time, there is not enough time to bring on-line a large coal-fired plant to make up a deficit in generation. Participants in the energy market, on the other hand, usually prefer a shorter gate closure because it reduces their exposure to risk. A load forecast calculated one hour ahead of real time is usually much more accurate than a forecast calculated four hours ahead. A retailer would therefore like to trade electronically up to the last minute to match its purchases with its anticipated load. This is considered preferable to relying on the managed spot market in which it is exposed to prices over which it has no control. Generators too prefer shorter gate closures because of the risk of sudden unit outage. If a unit fails after gate closure, there is nothing that the generator can do except hope that the spot market price will not be too high. On the other hand, if the unit fails before gate closure, the generator can try to make up the deficit in generation by purchasing at the best possible price on the electronic exchange. In general, traders prefer a true spot market that is driven solely by market forces to a managed spot market that is heavily influenced by complex technical considerations.

3.5.3 Operation of the managed spot market

Figure 3.2 summarizes the operation of a managed spot market. At gate closure, the producers and the consumers must inform the SO of their contractual positions, that is, how much power they intend to produce or consume during the period under consideration. The SO combines that information with its own forecast of the total load to determine by how much the system is likely to be in imbalance. If generation exceeds the load, the system is said to be long. If the opposite holds, the system is short. The SO must then decide which balancing bids and offers it will use to cover the imbalances.

When an electricity market operates on the basis of a pool run by the system operator, the balancing function will often be so closely integrated with the energy market function that they may be difficult to separate.

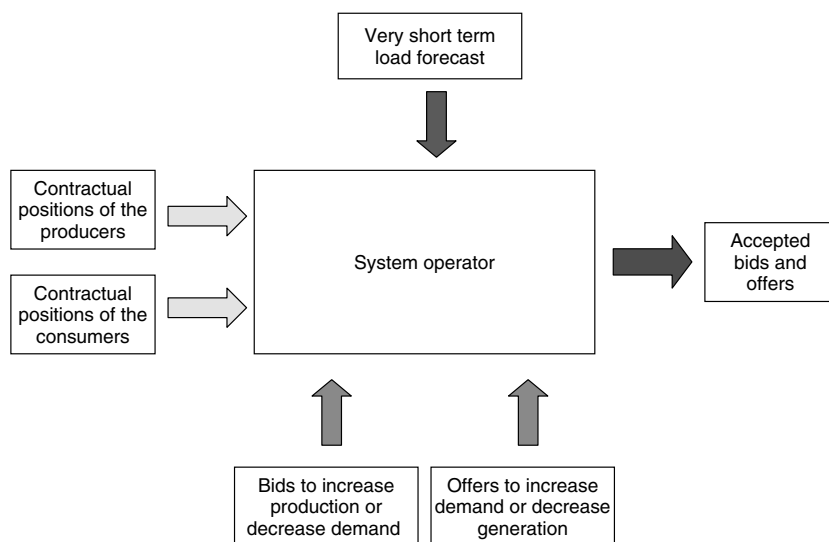


Figure 3.2 Schematic diagram of the operation of a managed spot market for electricity

3.5.3.1 Example 3.3

Let us revisit our trader for Borduria Power from Example 3.1 and see how she handles the spot market. When the gate closes for bilateral trading, Fiona has contracted to produce a net amount of 580 MWh for the period under consideration. She informs the system operator that her company intends to produce this amount as follows:

Unit	Scheduled production (MW)
A	500
B	80
C	0

Fiona then has to decide what bids and offers she wants to make in the managed spot market. To help her in this decision, she considers the scheduled output and the characteristics of Borduria Power's generating units:

Unit	p^{sched}	p^{min}	p^{max}	MC
	(MW)	(MW)	(MW)	(\$/MWh)
A	500	100	500	10.0
B	80	50	80	13.0
C	0	0	50	17.0

The only bid to increase generation that she can offer involves unit C because units A and B are scheduled to produce their maximum output. Such a bid would be for a maximum of 50 MWh and would have to be priced at 17.00 \$/MWh or above to be profitable, if we assume that the start-up cost of unit C is negligible.

Fiona also considers the possibility of reducing the production of units A and B. She would be willing to pay up to 10 \$/MWh to reduce the output of unit A and up to 13 \$/MWh to reduce the output of unit B because that is the marginal cost of producing power with these units. The output of these units can be reduced by 400 MW and 30 MW respectively without affecting the plans for the following periods, if we assume that there are no restrictions on the ramp rate of the units. Further reductions would require their shutdown and might then preclude Borduria Power from meeting its commitments for later hours. Furthermore, the cost of restarting these units would reduce their profitability.

In Chapter 2, we argued that, in a perfectly competitive market, the optimal strategy of each participant is to bid its marginal cost or to offer its marginal value. As we will discuss in a subsequent section, electricity markets are usually not perfectly competitive. Some participants can increase their profits by bidding above their marginal costs or offering below their marginal value. On the basis of her experience, Fiona decides that the following bids and offers are likely to maximize Borduria Power's profit:

Type	Identifier	Price (\$/MWh)	Amount (MW)
Bid	SMB-1	17.50	50
Offer	SMO-1	12.50	30
Offer	SMO-2	9.50	400

We will revisit this example one more time after discussing the settlement process.

3.5.4 Interactions between the managed spot market and the other markets

Since the managed spot market is the market of last resort for electrical energy, it has a strong influence on the other markets. If the spot price tends to be low, purchasers of energy will not be unduly concerned about being short because they can make up their deficits in the spot market at a reasonable price. They might therefore buy slightly less than they need in the forward market and will thus push down the price of energy in this market. On the other hand, if the spot price tends to be high, these same purchasers will push up the price in the forward market as they buy more to make sure that they cover all their needs at the better price. If electricity was a simple commodity, such discrepancies should even themselves out over time and the forward prices should reflect the expected value of the spot price.

Electricity is certainly not the only commodity whose spot price is highly volatile. A weather forecast predicting frost in the coffee-producing regions of Brazil will send

the spot price of coffee soaring. This price may very well come crashing down the next day if the forecast turns out to be inaccurate or the damage to the crop is more limited than was feared. The difference between coffee and electrical energy is that coffee traded in the spot market is produced exactly the same way as coffee sold under a long-term contract. On the other hand, an MWh sold through the managed spot market often will have been produced by a plant that is much more flexible than the plants that generate the bulk of the energy consumed at that time.

Ensuring the smooth and secure operation of a power system requires the performance of ancillary services such as load following and frequency response as well as the provision of spinning reserve capacity. Providing these services is another way for generators and consumers to increase their revenues. A few generating plants that are very flexible but find it hard to compete in the energy market usually occupy this market niche. The provision of these services translates into the delivery of a relatively small amount of electrical energy. If the providers of these services are remunerated on the basis of the megawatt-hour produced, they will have to charge a fairly high price per megawatt-hour to collect enough revenue to stay in business. Including the cost of this energy in the calculation of the spot price will often result in sharp price spikes. These price spikes do not reflect a sudden deficit of electrical energy in the market. They are a consequence of an acute but temporary lack of liquidity. Price spikes occur because, for a short time, the number of participants able to provide this “service energy” is very small and because consumers are not able or not willing to reduce their demand at short notice. Price spikes represent a risk for companies that are forced to buy from the managed spot market and will encourage them to purchase more in the forward markets and hence drive up the forward prices. These forward prices are thus artificially inflated by the need to generate a small fraction of the total energy demand at short notice. Chapter 5 discusses in more detail the provision of ancillary services and alternative methods of remunerating the participants that provide them.

3.6 The Settlement Process

Commercial transactions are normally settled directly between the two parties involved: following the delivery of the goods by the seller to the buyer, the buyer pays the seller the agreed price. If the amount delivered is less than the amount contracted, the buyer is entitled to withhold part of the payment. Similarly, if the buyer consumes more than the agreed amount, the seller is entitled to an additional payment. This process is more complex for electricity markets because the energy is pooled during its transmission from the producers to the consumers. This is why a centralized settlement system is needed.

For bilateral transactions in electrical energy, the buyer pays the seller the agreed price as if the agreed quantity had been delivered exactly. Similarly, the anonymous transactions arranged through screen-based trading are settled through the intermediary of the power exchange as if they had been executed perfectly. However, there will always be inaccuracies in the completion of the contracts. If a generator fails to produce the amount of energy that it has contracted to sell, the deficit cannot simply be withheld from this generator’s customers. Instead, to maintain the stability of the

system, the system operator buys replacement energy on the managed spot market. Similarly, if a large user or retailer consumes less than it has bought, the system operator sells the excess on the managed spot market. These balancing activities make all bilateral contracts look as if they have been fulfilled perfectly. They also carry a cost. In most cases, the amount of money paid by the system operator to purchase replacement energy is not equal to the amount of money earned when selling excess energy. The parties that are responsible for the imbalances should pay the cost of these balancing activities.

The first step in the settlement process consists, therefore, in determining the net position of every market participant. To this end, each generator must report to the settlement system the net amount of energy that it had contracted to sell for each period, including the energy traded through the managed spot market. This amount is subtracted from the amount of energy that it actually produced. If the result is positive, the generator is deemed to have sold this excess energy to the system. On the other hand, if the result is negative, the generator is treated as if it had bought the difference from the system.

Similarly, all large consumers and retailers must report the net amount of energy that they had contracted to buy for each period, including the energy traded through the managed spot market. This amount is subtracted from the amount of energy actually consumed. Depending on the sign of the result, the consumer or the retailer is deemed to have sold energy to the system or bought energy from the system.

These imbalances are charged at the spot market price. If this market is suitably competitive, this price should reflect the incremental cost of balancing energy. As we discussed in the previous section and as we will investigate further in Chapter 5, it is debatable whether the cost of the energy supplied by participants providing ancillary services should be included in this price.

Settlement in a pool-based electricity market is more straightforward because all transactions take place through the pool.

3.6.1.1 Example 3.4

In Examples 3.1 and 3.3, we looked at the trading activities of Borduria Power for the period from 2:00 to 3:00 P.M. on 11 June in the bilateral market and the managed spot market. Let us assume that the following events took place after gate closure:

- Faced with a deficit of generation, the System Operator called 40 MWh of Borduria Power's spot market bid at 17.50 \$/MWh (SMB-1).
- The troubles with Borduria Power's unit B turned out to be worse than anticipated, forcing its complete shutdown soon after the beginning of the period. It was only able to produce 10 of the 80 MWh that it was scheduled to produce, leaving Borduria Power with a deficit of 70 MWh.
- The spot price of electrical energy was 18.25 \$/MWh for this period.

The following table shows the details of the flows of money in and out of Borduria Power's trading account.

Market	Identifier	Amount (MWh)	Price (\$/MWh)	Income (\$)	Expense (\$)
Futures and forwards	Cheapo Energy	200	12.50	2500.00	
	Borduria Steel	250	12.80	3200.00	
	Quality Electrons	100	14.00	1400.00	
	Perfect Power	−30	13.50		405.00
	Cheapo Energy	50	13.80	690.00	
Power exchange	O1	20	13.50	270.00	
	O2	30	13.30	399.00	
	O3	10	13.25	132.50	
	B3	−20	14.40		288.00
	B6	−20	14.30		286.00
	B8	−10	14.10		141.00
Spot market	SMB-1	40	17.50	700.00	
	Imbalance	−70	18.25		1277.50
Total		550		9291.50	2397.50

Bilateral trades are settled directly between Borduria Power and its counterparties. Since trades on the power exchange are anonymous, they are settled through BPex (the company running the power exchange). Finally, activity on the managed spot market (both voluntary and compulsory) is settled through the system operator or its settlement agent. The bottom line of this table indicates that the trading revenue of Borduria Power for this period amount to \$6894.00. To determine whether this trading period was profitable, we would have to compute the cost of producing the energy that Borduria Power delivered. Carrying out this computation for a single trading period is rather difficult because there is no simple way to allocate the start-up and no-load costs of the generating units.

3.7 Further Reading

The theory of spot pricing was first applied to electrical energy by Schweppe *et al.* (1988) and is described in detail in Schweppe's book that is often credited with providing the theoretical underpinning for the introduction of competition in power systems. The book by Stoft (2002) discusses the design of electricity markets in much more detail than here. A considerable amount of material on the organization of electricity markets is available from regulatory bodies such as the Federal Energy Regulatory Commission (FERC) in the United States and the Office of Gas and Electricity Markets (OFGEM) in the United Kingdom or from market operators such as PJM.

FERC, <http://www.ferc.gov>.

OFGEM, <http://www.ofgem.gov.uk/public/adownloads.htm#retabm>.

PJM, <http://www.pjm.com>.

Schweppe F C, Caramanis M C, Tabors R D, Bohn R E, *Spot Pricing of Electricity*, Kluwer Academic Publishers, Boston, MA 1988.

Stoft S, *Power System Economics*, John Wiley & Sons, 2002.

3.8 Problems

- 3.1 Choose an electricity market about which you have access to sufficient information, preferably the same market that you studied for the problems of Chapter 1. Describe the implementation of this market. In particular, determine the aspects that are based on bilateral trading and those that are centrally operated. Discuss the mechanism used to set prices in the managed spot market.
- 3.2 The rules of the Syldavian electricity market stipulate that all participants must trade energy exclusively through the Power Pool. However, the Syldavia Aluminum Company (SALCo) and the Northern Syldavia Power Company (NSPCo) have signed a contract for difference for the delivery of 200 MW on a continuous basis at a strike price of 16 \$/MWh.
 - a. Trace the flow of power and money between these companies when the pool price takes the following values: 16 \$/MWh, 18 \$/MWh and 13 \$/MWh.
 - b. What happens if during one hour the Northern Syldavia Power Company is able to deliver only 50 MWh and the pool price is 18 \$/MWh?
 - c. What happens if during one hour the Syldavia Aluminum Company consumes only 100 MWh and the pool price is 13 \$/MWh?
- 3.3 The following six companies participate, along with others, in the Southern Antarctica electrical energy market:
 - **Red:** A generating company owning a portfolio of plants with a maximum capacity of 1000 MW.
 - **Green:** Another generating company with a portfolio of plants with a maximum capacity of 800 MW.
 - **Blue:** A retailer of electrical energy.
 - **Yellow:** Another retailer of electrical energy.
 - **Magenta:** A trading company with no generating assets and no demand.
 - **Purple:** Another trading company with no physical assets.

The following information pertains to the operation of this market for Monday 29 February 2016 between 1:00 and 2:00 P.M.

Load forecasts Blue and Yellow forecast that their customers will consume 1200 MW and 900 MW respectively during that hour.

Long-term contracts June 2015: Red signs a contract for the supply of 600 MW at 15 \$/MWh for all hours between 1 January 2015 and 31 December 2020

July 2015: Blue signs a contract for the purchase of 700 MW for all hours between 1 February 2016 and 31 December 2016. The price is set at 12 \$/MWh for off-peak hours and at 15.50 \$/MWh for peak hours.

August 2015: Green signs a contract for the supply of 500 MW at 16 \$/MWh for peak hours in February 2016.

September 2015: Yellow signs a contract for the purchase of electrical energy. The contract specifies a profile of daily and weekly volumes and a profile for daily and weekly prices. In particular, on weekdays between 1 : 00 and 2 : 00 P.M., the volume purchased is 550 MW at 16.25 \$/MWh.

Futures contracts All contracts are for delivery on 29 February 2016 between 1 : 00 and 2 : 00 P.M.

Date	Company	Type	Amount	Price
10/9/15	Magenta	Buy	50	14.50
20/9/15	Purple	Sell	100	14.75
30/9/15	Yellow	Buy	200	15.00
10/10/15	Magenta	Buy	100	15.00
20/10/15	Red	Sell	200	14.75
30/10/15	Green	Sell	250	15.75
30/10/15	Blue	Buy	250	15.75
10/11/15	Purple	Buy	50	15.00
15/11/15	Magenta	Sell	100	15.25
20/11/15	Yellow	Buy	200	14.75
30/11/15	Blue	Buy	300	15.00
10/12/15	Red	Sell	200	16.00
15/12/15	Red	Sell	200	15.50
20/12/15	Blue	Sell	50	15.50
15/1/16	Purple	Sell	200	14.50
20/1/16	Magenta	Buy	50	14.25
10/2/16	Yellow	Buy	50	14.50
20/2/16	Red	Buy	200	16.00

25/2/16	Magenta	Sell	100	17.00
28/2/16	Purple	Buy	250	14.00
28/2/16	Yellow	Sell	100	14.00

Options contracts In November 2015, Red bought a put option for 200 MWh at 14.75 \$/MWh. The option fee was \$50.

In December 2015, Yellow bought a call option for 100 MWh at 15.50 \$/MWh. The option fee was \$25.

Outcome

- The spot price on the Southern Antarctica electricity market was set at 15.75 \$/MWh for 29 February 2016 between 1:00 and 2:00 P.M.
- Owing to the difficulties at one of its major plants, Red was able to generate only 800 MW. Its average cost of production was 14.00 \$/MWh.
- Green generated 770 MW at an average cost of 14.25 \$/MWh.
- Blue's demand turned out to be 1250 MW. Its average retail price was 16.50 \$/MWh.
- Yellow's demand turned out to be 850 MW. Its average retail price was 16.40 \$/MWh.

Assuming that all imbalances are settled at the spot market price, calculate the profit or loss made by each of these participants.

- 3.4 The operator of a centralized market for electrical energy has received the bids shown in the table below for the supply of electrical energy during a given period.

Company	Amount (MWh)	Price (\$/MWh)
Red	100	12.5
Red	100	14.0
Red	50	18.0
Blue	200	10.5
Blue	200	13.0
Blue	100	15.0
Green	50	13.5
Green	50	14.5
Green	50	15.5

- a. Build the supply curve
- b. Assume that this market operates unilaterally, that is, that the demand does not bid and is represented by a forecast. Calculate the market price, the quantity produced by each company and the revenue of each company for each of the following loads: 400 MW, 600 MW, 875 MW.
- c. Suppose that instead of being treated as constant, the load is represented by its inverse demand curve, which is assumed to have the following form:

$$D = L - 4.0 \cdot \pi$$

where D is the demand, L is the forecasted load and π is the price. Calculate the effect that this price sensitivity of demand has on the market price and the quantity traded.

- 3.5 The Syldavian Power and Light Company owns one generating plant and serves some load. It has been actively trading in the electricity market and has established the following position for 11 June between 10:00 and 11:00 A.M.:

- Long-term contract for the purchase of 600 MW during peak hours at a price of 20.00 \$/MWh
- Long-term contract for the purchase of 400 MW during off-peak hours at a price of 16.00 \$/MWh
- Long-term contract with a major industrial user for the sale of 50 MW at a flat rate of 19.00 \$/MWh
- The remaining customers purchase their electricity at a tariff of 21.75 \$/MWh
- Future contract for the sale of 200 MWh at 21.00 \$/MWh
- Future contract for the purchase of 100 MWh at 22.00 \$/MWh
- Call option for 150 MWh at an exercise price of 20.50 \$/MWh
- Put option for 200 MWh at an exercise price of 23.50 \$/MWh.
- Call option for 300 MWh at an exercise price of 24.00 \$/MWh.

The option fee for all the options is 1.00 \$/MWh. The peak hours are defined as being the hours between 8:00 A.M. and 8:00 P.M.

The outcome for 11 June between 10:00 and 11:00 is as follows:

- The spot price is set at 21.50 \$/MWh.
 - The total load of the Syldavian Power and Light Company is 1200 MW, including the large industrial customer.
 - The power plant produces 300 MWh at an average cost of 21.25 \$/MWh.
- a. Assuming that all imbalances are settled at the spot market price, calculate the profit or loss made by the company during that hour.
 - b. What value of the spot market would reduce the profit or loss of the company to zero? Would this change in spot price affect any of the option contracts?

- 3.6 A company called Borduria Energy owns a nuclear power plant and a gas-fired power plant. Its trading division has entered into the following contracts for 25 January:

T-1. A forward contract for the sale of 50 MW at a price of 21.00 \$/MWh. This contract applies to all hours.

T-2. A long-term contract for the sale of 300 MW during off-peak hours at a price of 14.00 \$/MWh

T-3. A long-term contract for the sale of 350 MW at 20 \$/MWh during peak hours.

In addition, for the trading period from 2:00 to 3:00 P.M. on that day, it has entered into the following transactions:

T-4. A future contract for the purchase of 600 MWh at 20.00 \$/MWh

T-5. A future contract for the sale of 100 MWh at 22.00 \$/MWh

T-6. A put option for 250 MWh at an exercise price of 23.50 \$/MWh

T-7. A call option for 200 MWh at an exercise price of 22.50 \$/MWh

T-8. A put option for 100 MWh at an exercise price of 18.75 \$/MWh

T-9. A bid in the spot market to produce 50 MW using its gas-fired plant at 19.00 \$/MWh

T-10. A bid in the spot market to produce 100 MW using its gas-fired plant at 22.00 \$/MWh

The option fee for all call and put options is \$2.00/MWh. The peak hours are defined as being the hours between 8:00 A.M. and 8:00 P.M.

Borduria Energy also sells electrical energy directly to small consumers through its retail division. Residential customers pay a tariff of 25.50 \$/MWh

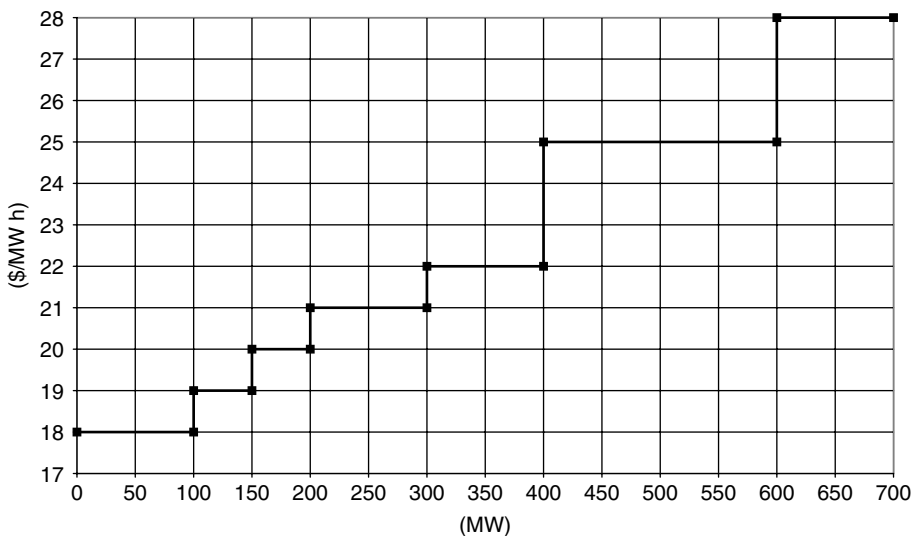


Figure 3.3 Stack of bids for Problem 3.6

and commercial consumers pay a tariff of 25.00 \$/MWh. Borduria Energy does not sell electricity to industrial consumers.

The graph on Figure 3.3 shows the stack of bids that the spot market operator has received for the trading period from 2 : 00 to 3 : 00 P.M. on 25 January. In order to balance load and generation, it accepted bids for 225 MW in increasing order of price for that hour. The spot price was set at the price of the last accepted bid.

During that hour, the residential customers served by Borduria Energy consumed 300 MW, while its commercial customers consumed 200 MW. The nuclear power plant produced 400 MWh at an average cost of 16.00 \$/MWh. Its gas-fired plant produced 200 MWh at an average cost of 18.00 \$/MWh. All imbalances are settled at the spot market price.

- a. Calculate the profit or loss made by Borduria Energy during that hour.
- b. Calculate the effect that the sudden outage of the nuclear generating plant at 2 : 00 P.M. on 25 January would have on the profit (or loss) of Borduria Energy for that hour.

4

Participating in Markets for Electrical Energy

4.1 Introduction

In the previous chapter, we discussed the basic principles of markets for electrical energy and illustrated through some examples how market participants interact with such markets. In this chapter, we discuss in more detail the decisions that generators, consumers and others take to optimize the benefits that they derive from these markets.

We will first discuss why consumers have a much more passive role than producers do in electricity markets and how retailers serve as their intermediaries in the market for electrical energy.

We will then adopt the perspective of a generating company and consider the case in which this company faces a perfectly competitive market. In such a market, since the company's actions do not affect the prices, it can optimize its activities independently of what other producers or consumers might do. Such a scenario is unrealistic in the context of electricity markets because the short-term elasticity of the demand for electricity is very low and because in most markets the bulk of the electrical energy is produced by a small number of producers. We will therefore discuss some of the techniques that have been proposed to analyze the operation of imperfectly competitive markets and to maximize a producer's profit in such markets.

Finally, we will explore how storage facilities and other hybrid participants can make a profit from trading electrical energy.

4.2 The Consumer's Perspective

Microeconomic theory suggests that consumers of electricity, like consumers of all other commodities, increase their demand up to the point at which the marginal benefit they derive from the electricity is equal to the price they have to pay. For example, a manufacturer will not produce widgets if the cost of the electrical energy required to produce them makes their sale unprofitable. Similarly, the owner of a fashion boutique will increase the lighting level only up to the point at which it attracts more customers. Finally, at home during a cold winter evening, there comes a point at which most of us

will put on some extra clothes rather than turning up the thermostat and face a very large electricity bill. Since this chapter deals only with the short-term behavior of consumers, we do not consider the possibility that they will purchase new appliances, machinery or other facilities that would allow them to change their pattern of consumption.

If these industrial, commercial and residential customers pay a flat rate for each kilowatt-hour that they consume, they are insulated from the spot price of electricity and their demand is affected only by the cycle of their activities. Averaged over a few weeks or months, their demand reflects only their willingness to pay this flat rate. But what happens when the price of electrical energy fluctuates more rapidly? Empirical evidence suggests that demand does decrease in response to a short-term price increase, but that this effect is relatively small. In other words, the price elasticity of the demand for electricity is small. On a price versus quantity diagram, the slope of the demand curve is therefore very steep. Determining the shape of the demand curve with any kind of accuracy is practically impossible for a commodity like electrical energy. It is, however, interesting to compare the range of prices for electrical energy sold on a competitive market such as the Electricity Pool of England and Wales (see Table 4.1) with a measure of the value that consumers place on the availability of electrical energy. One such measure is the value of lost load (VOLL), which is obtained through surveys of consumers and represents the average price per megawatt-hour that consumers would be willing to pay to avoid being disconnected without notice. For the same period as the data shown in Table 4.1, the VOLL used in England and Wales was set at 2768 £/MWh.

Two economic and social factors explain this weak elasticity. First, the cost of electrical energy makes up only a small portion of the total cost of producing most industrial goods and represents only a small fraction of the cost of living for most households. At the same time, electricity is indispensable in manufacturing and most individuals in the industrialized world regard it as essential to their quality of life. Most industrial consumers therefore will not reduce their production drastically to avoid a small increase in their electricity costs. In the short term, the savings might be more than offset by the loss of revenue. Similarly, most residential consumers will probably not reduce their comfort and convenience to cut their electricity bill by a few percents. The second factor explaining this weak elasticity is historical. Since the early days of commercial electricity generation over a century ago, electricity has been marketed as a commodity that is easy to use and always available. This convenience has become so ingrained that it is fair to say that very few people carry out a cost/benefit analysis each time they turn on the light!

Rather than simply reducing their demand in response to a sudden increase in the price of electrical energy, consumers instead may decide to delay this demand until a time when prices are lower. For example, a manufacturer may decide to delay the

Table 4.1 Pool selling price on the electricity pool of England and Wales (in £/MWh)

	Minimum	Maximum	Average
January 2001	0.00	168.49	21.58
February 2001	10.00	58.84	18.96
March 2001	8.00	96.99	20.00

(Source: The Electricity Pool of England and Wales).

completion of a particularly energy-intensive step of a production process until the night shift if the price of electrical energy is expected to be lower at that time. Similarly, residential consumers in some countries take advantage of lower nighttime tariffs by waiting until later in the evening to wash and dry clothes or heat water. Shifting demand is possible only if the consumer is able to store intermediate products, heat, electrical energy, or dirty clothes. Such storage facilities and the associated control equipment carry a significant investment cost. The savings that can be achieved by shifting demand for electricity from periods of high prices to periods of low prices might not justify these costs. Furthermore, managing demand requires more flexibility or more willingness to accept a loss of convenience than may be available.

Most small residential and commercial consumers therefore will not be very interested in reacting to hourly or half-hourly price changes. Even if they were, the cost of the communication infrastructure needed to inform them of these prices and to register their consumption during each period would absorb most, if not all, of the potential benefits. For the foreseeable future, these consumers probably will continue purchasing electrical energy on the basis of a tariff. Such tariffs insulate them from daily fluctuations in prices and therefore reduce to zero their contribution to the overall short-term elasticity of demand.

This very low elasticity of the demand has undesirable effects on the operation of markets for electrical energy. In particular, when we discuss imperfect markets, we will see that it facilitates the exercise of market power by the producers.

4.2.1 Retailers of electrical energy

Consumers whose peak demand is at least a few hundreds kilowatts may be able to save significant amounts of money by employing specialized personnel to forecast their demand and trade in the electricity markets to obtain lower prices. Such consumers can be expected to participate directly and actively in the markets. On the other hand, such active trading is not worthwhile for smaller consumers. These smaller consumers usually prefer purchasing on a tariff, that is, at a constant price per kilowatt-hour that is adjusted at most a few times per year. Electricity retailers are in business to bridge the gap between the wholesale market and these smaller consumers.

The challenge for them is that they have to buy energy at a variable price on the wholesale market and sell it at a fixed price at the retail level. A retailer will typically lose money during periods of high prices because the price it has to pay for energy is higher than the price at which it resells this energy. On the other hand, during periods of low prices it makes a profit because its selling price is higher than its purchase price. To stay in business, the quantity-weighted average price at which a retailer purchases electrical energy should therefore be lower than the rate it charges its customers. This is not always easy to achieve because the retailer does not have direct control over the amount of energy that its customers consume. Each retailer is deemed to have sold to its customers the amount of energy that went through their meters. If for any period the aggregate amount over all its customers exceeds the amount that it has contracted to buy, the retailer has to purchase the difference on the spot market at whatever value the spot price reached for that period. Similarly, if the amount contracted exceeds the amount consumed by its customers, the retailer is deemed to have sold the difference on the spot market.

To reduce its exposure to the risk associated with the unpredictability of the spot market prices, a retailer therefore tries to forecast as accurately as possible the demand of its customers. It then purchases energy on the various markets to match this forecast. A retailer thus has a strong incentive to understand the consumption patterns of its customers. It will often encourage its customers to install meters that record the energy consumed during each period so it can offer them more attractive tariffs if they reduce their energy consumption during peak price hours. By taking into account all the meteorological, astronomical, economic, cultural and special factors that influence the consumption of electricity and using the most sophisticated forecasting techniques available, it is possible to predict the value of the demand at any hour with an average accuracy of about 1.5 to 2%. However, such accuracy is possible only with large groups of consumers in which the aggregation effects reduce the relative importance of the random fluctuations. A retailer who does not have a monopoly on the supply of electricity in a given region can therefore forecast the demand of its customers with considerably less accuracy than what a monopoly utility can achieve. This problem is exacerbated if, as one would expect, customers have the opportunity to change retailer to get a better tariff. An unstable customer base makes it much harder for the retailer to gather the reliable statistical data it needs to refine its demand forecast.

4.2.1.1 Example 4.1

Table 4.2 illustrates the daily operation of a retailer. Figures 4.1, 4.2 and 4.3 give a graphical representation of the data contained in this table. As shown on the second and third lines of Table 4.2, our retailer has forecasted the demand of its customers for a 12-h period and has purchased energy to meet this anticipated demand. The amount purchased for each hour results from a combination of contracts of various types (long-term bilateral, forwards, futures, screen-based transactions). The fourth and fifth lines of the table show respectively the average and total costs of the energy purchased for each period. The average cost tends to be higher during hours of peak demand.

As one might expect, the actual demand does not match the forecast and there are positive and negative imbalances at each hour. These imbalances are compulsorily settled at the spot prices shown in the eighth line of the table and result in additional balancing costs (or revenue if the imbalance is negative) for our retailer. Adding the balancing and contract costs gives the total cost of energy for each hour. We will assume that our retailer has opted for a very simple tariff structure and charges a flat rate of 38.50 \$/MWh to all of its customers. The “Total Revenues” and “Profits” lines of the table show the amounts that accrue for each hour. Our retailer makes an operational profit during the hours of low prices and a loss during the hours of high prices. Overall, for this 12-h period, the bottom line shows a loss of \$1154. Our retailer has to hope that this is not a typical period and that the average cost of purchasing electricity will be lower on other days. If this turns out to be a typical day, the retail rate will have to be raised to above the average purchasing cost of electrical energy (including spot purchases), which is \$39.23 for this period. The relatively high balancing costs suggest that our retailer could also increase its profitability by

Table 4.2 Data for Example 4.1

Period	Units	1	2	3	4	5	6	7	8	9	10	11	12	Average	Total
Load forecast	(MWh)	221	219	254	318	358	370	390	410	382	345	305	256	325	3828
Contract purchases	(MWh)	221	219	254	318	358	370	390	410	382	345	305	256	325	3828
Average costs	(\$/MWh)	24.70	24.5	27.50	35.20	40.70	42.40	45.50	48.60	44.20	38.80	33.40	27.70	36.10	
Contract costs	(\$)	5459	5366	6985	11194	14571	15688	17745	19926	16884	13386	10187	7091	12040	144482
Actual loads	(MWh)	203	203	287	328	361	401	415	407	397	381	331	240	330	3954
Imbalances	(MWh)	-18	-16	33	10	3	31	25	-3	15	36	26	-16	10.5	
Spot prices	(\$/MWh)	13.20	12.50	17.40	33.30	69.70	75.40	70.10	102.30	81.40	63.70	46.90	18.30	50.35	
Balancing costs	(\$)	-238	-200	574	333	209	2337	1753	-307	1221	2293	1219	-293	742	8901
Total costs	(\$)	5221	5166	7559	11527	14780	18025	19498	19619	18105	15679	11406	6798	12782	153383
Total revenues	(\$)	7815.5	7815.5	11050	12628	13899	15439	15978	15670	15285	14669	12744	9240	12686	152229
Profits	(\$)	2595	2650	3491	1101	-882	-2587	-3521	-3950	-2821	-1011	1338	2442	-96	-1154
Profits w/o error	(\$)	3050	3066	2794	1049	-788	-1443	-2730	-4141	-2177	-104	1556	2765	241	2896

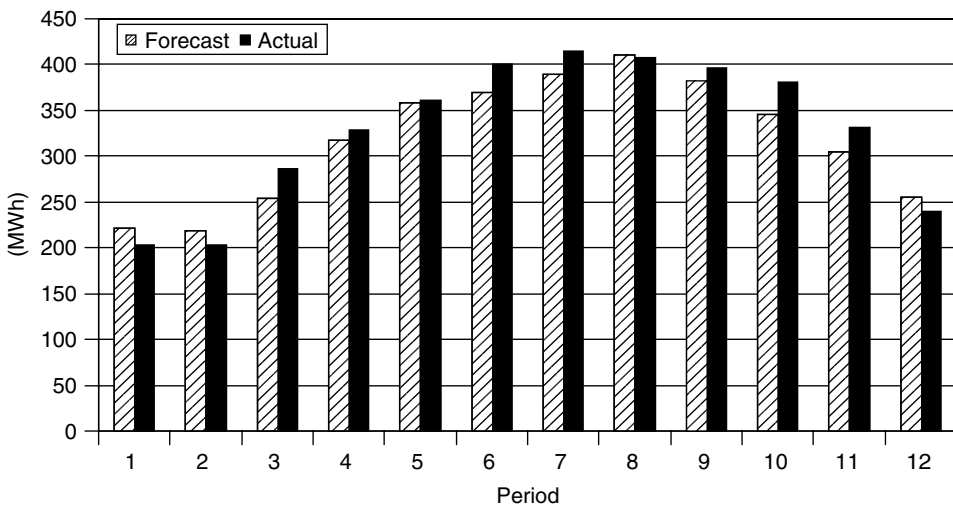


Figure 4.1 Forecast and actual demand for Example 4.1

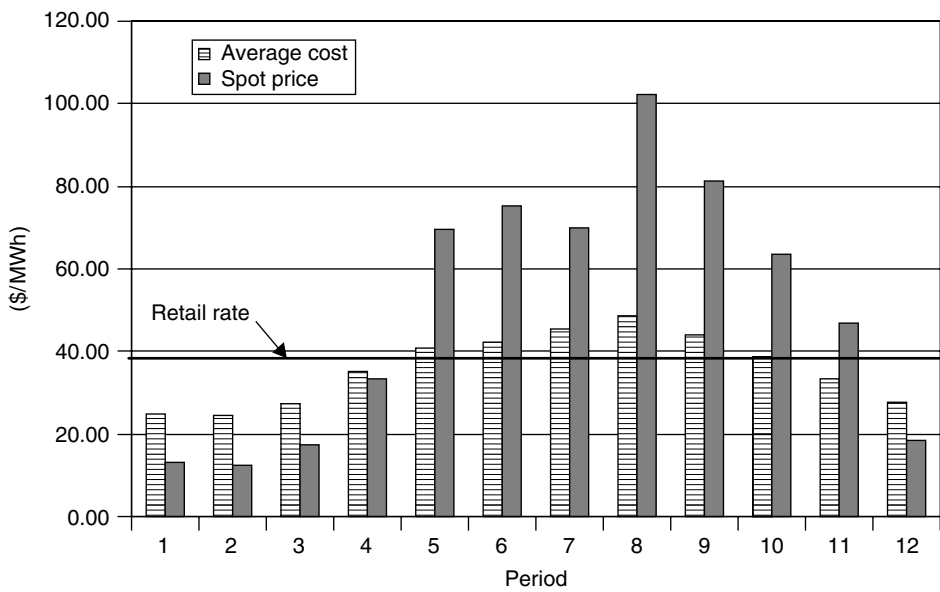


Figure 4.2 Costs and prices for Example 4.1

improving the accuracy of its forecast. To illustrate this point, the last line of the table shows what the profits would be if the demand turned out to be exactly equal to the forecast and the retailer was not exposed to the spot prices. If this perfect forecast had been achieved during this period, our retailer would have made a profit of \$2896.

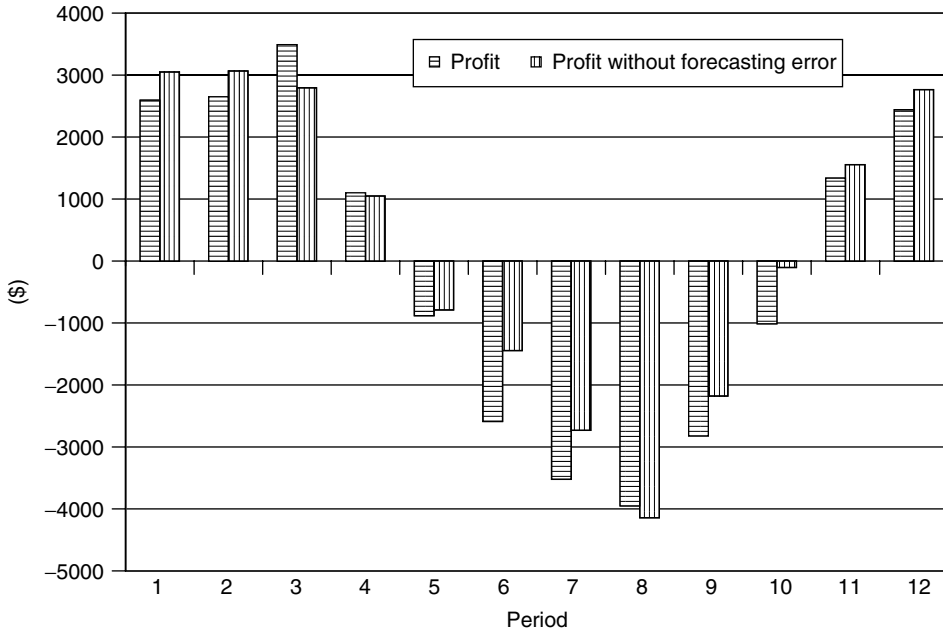


Figure 4.3 Profits and losses for Example 4.1

4.3 The Producer's Perspective

In this section, we will take the perspective of a generating company that tries to maximize the profits it derives from the sale of electrical energy produced by a single generating unit called unit i . For the sake of simplicity, we will consider a period of one hour and we will assume that all quantities remain constant during that period. The maximization of the profit from this unit during this hour can be expressed as the difference between the revenue resulting from the sale of the energy it produces and the cost of producing this energy:

$$\max \Omega_i = \max[\pi \cdot P_i - C_i(P_i)] \quad (4.1)$$

where P_i is the power produced by unit i during that hour, π is the price at which this energy is sold and $C_i(P_i)$ is the cost of producing this energy. If we assume that the only variable over which the company has direct control is the power produced by this unit, the necessary condition for optimality corresponding to Equation (4.1) is

$$\frac{d\Omega_i}{dP_i} = \frac{d(\pi \cdot P_i)}{dP_i} - \frac{dC_i(P_i)}{dP_i} = 0 \quad (4.2)$$

The first term in this expression represents the marginal revenue of unit i , that is, the revenue the company would get for producing an extra megawatt during this hour. The

second term represents the cost of producing this extra megawatt, that is, the marginal cost. To maximize profits, the production of unit i must therefore be adjusted up to the level at which its marginal revenue is equal to its marginal cost:

$$MR_i = MC_i \quad (4.3)$$

4.3.1 Perfect competition

4.3.1.1 Basic dispatch

If competition is perfect (or if the potential output of the unit is very small compared to the size of the market), the price π is not affected by changes in P_i . The marginal revenue of unit i is thus

$$MR_i = \frac{d(\pi \cdot P_i)}{dP_i} = \pi \quad (4.4)$$

which expresses the fact that a price-taking generator collects the market price for each megawatt-hour that it sells. Under these conditions, if the marginal cost is a monotonically increasing function of the power produced, the generating unit should increase its output up to the point at which the marginal cost of production is equal to the market price:

$$\frac{dC_i(P_i)}{dP_i} = \pi \quad (4.5)$$

The marginal cost includes the costs of fuel, maintenance and all other items that vary with the power produced by the unit. Costs that are not a function of the amount of power produced in the period under consideration (e.g. the amortized cost of building the plant or the fixed maintenance and personnel costs) are not factored in the marginal cost and are thus irrelevant when making short-term production decisions.

As long as competition is perfect, the output of each generating unit should be determined using Equation (4.5). Since the price is considered as given, this implies that all generating units can be dispatched independently, even if a generating company owns more than one unit. We will consider in a later section the much more complicated case in which the total capacity of the generating units owned by a single company is large enough to influence the price of energy.

4.3.1.2 Example 4.2

Generating units fired with fossil fuels are characterized by input–output curves that specify the amount of fuel (usually expressed in MJ/h or MBTU/h) required to produce a given and constant electrical power output for one hour.

Consider a coal-fired steam unit whose minimum stable generation is 100 MW (i.e. the minimum amount of power that it can produce continuously) and whose maximum output is 500 MW. On the basis of measurements taken at the plant, the input–output curve of this unit is estimated as

$$H_1(P_1) = 110 + 8.2 P_1 + 0.002 P_1^2 \text{ MJ/h}$$

The hourly cost of operating this unit is obtained by multiplying the input–output curve by the cost of fuel F in \$/MJ:

$$C_1(P_1) = 110 F + 8.2 F P_1 + 0.002 F P_1^2 \text{ \$/h}$$

If we assume that the cost of coal is 1.3 \$/MJ, the cost curve of this unit is

$$C_1(P_1) = 143 + 10.66 P_1 + 0.0026 P_1^2 \text{ \$/h}$$

If the price at which electrical energy can be sold is 12 \$/MWh, the output that this unit should produce is given by

$$\frac{dC_1(P_1)}{dP_1} = 10.66 + 0.0052 P_1 = 12 \text{ \$/MWh or } P_1 = 257.7 \text{ MW}$$

In practice, optimally dispatching even a single generating unit is more complex than Equation (4.5) suggests. In the following subsections, we will examine how the cost and technical characteristics of the generating units affect the basic dispatch.

4.3.1.3 Unit limits

Suppose that the maximum power $P_i^{\max}$ that can be produced by generating unit i is such that

$$\left. \frac{dC_i(P_i)}{dP_i} \right|_{P_i^{\max}} \leq \pi \quad (4.6)$$

This generating unit should therefore produce $P_i^{\max}$. On the other hand, if the minimum stable generation of unit i is such that

$$\left. \frac{dC_i(P_i)}{dP_i} \right|_{P_i^{\min}} > \pi \quad (4.7)$$

this unit cannot generate profitably at that price and the only way to avoid losing money on its operation is to shut it down.

4.3.1.4 Example 4.3

The generating unit of the previous example should operate at its maximum output whenever the price of electrical energy is greater than or equal to

$$\left. \frac{dC_i(P_i)}{dP_i} \right|_{500 \text{ MW}} = 10.66 + 0.0052 \cdot 500 = 13.26 \text{ \$/MWh}$$

On the other hand, this unit cannot operate profitably if the price drops below

$$\left. \frac{dC_i(P_i)}{dP_i} \right|_{100 \text{ MW}} = 10.66 + 0.0052 \cdot 100 = 11.18 \text{ \$/MWh}$$

4.3.1.5 Piecewise linear cost curves

Input–output curves are drawn on the basis of measurements taken while the generating unit is operating at various levels of output. Even if every effort is made to make these measurements as accurate as possible, the data points usually do not line up along a smooth curve. A piecewise linear interpolation of this data is therefore just as acceptable as a quadratic function.

Figure 4.4 shows a piecewise linear cost curve and its associated marginal cost curve. Since each segment of the cost curve is linear, each segment of the marginal cost curve is constant. This makes the process of dispatching the unit in response to

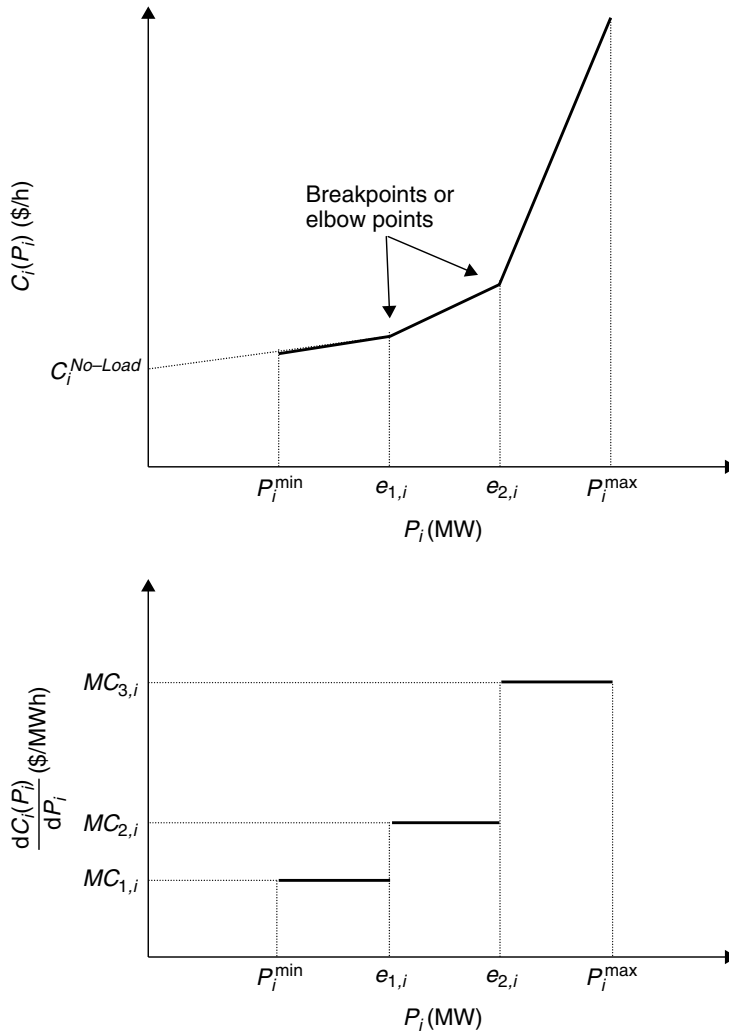


Figure 4.4 Piecewise linear cost curve and its associated piecewise constant incremental cost curve

electrical energy prices very simple.

$$\begin{aligned}
 \pi < MC_{1,i} &\Rightarrow P_i = P_i^{\min} \\
 MC_{1,i} < \pi < MC_{2,i} &\Rightarrow P_i = e_{1,i} \\
 MC_{2,i} < \pi < MC_{3,i} &\Rightarrow P_i = e_{2,i} \\
 MC_{3,i} < \pi &\Rightarrow P_i = P_i^{\max}
 \end{aligned} \tag{4.8}$$

If the price is exactly equal to the value of one of the segments of the marginal cost curve, the generation can take any value within that segment. The marginal cost at a breakpoint is equal to the slope of the next segment because the marginal cost is traditionally defined as the cost of the next megawatt, not the cost of the previous megawatt.

4.3.1.6 Example 4.4

The quadratic cost curve of Example 4.2 can be approximated by the following three-segment piecewise linear cost curve:

$$\begin{aligned}
 100 \leq P_1 \leq 250 : C_1(P_1) &= 11.57 P_1 + 78.0 \text{ \$/h} \\
 250 \leq P_1 \leq 400 : C_1(P_1) &= 12.35 P_1 - 117.0 \text{ \$/h} \\
 400 \leq P_1 \leq 500 : C_1(P_1) &= 13.00 P_1 - 377.0 \text{ \$/h}
 \end{aligned}$$

Figure 4.5 shows how this unit should be dispatched as the price paid for electrical energy varies.

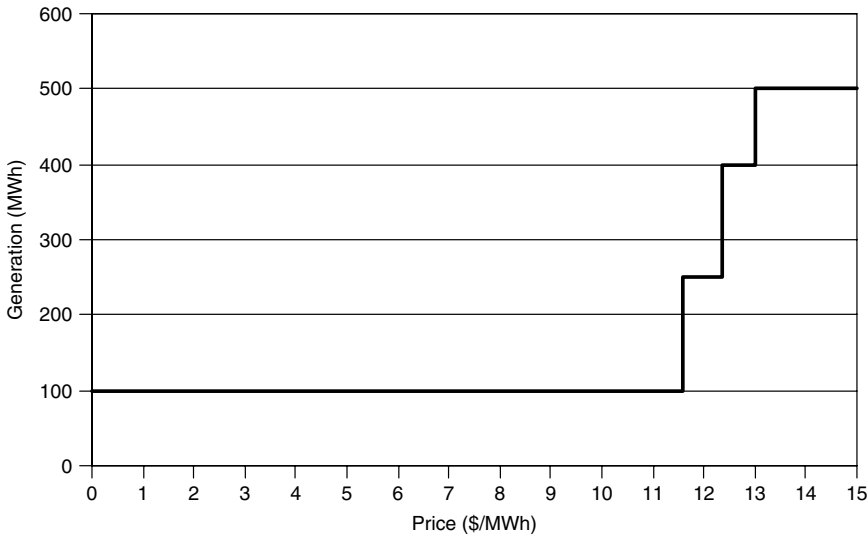


Figure 4.5 Dispatch of the generating unit of Example 4.4 as a function of the price of electrical energy

4.3.1.7 No-load cost

Producers do not decide the amount of electrical energy that they sell solely on the basis of a comparison between the market price and the marginal cost of production. Running a unit in such a way that its marginal cost is equal to the market price does not guarantee that a profit will be made. Producers must also consider the quasi-fixed costs associated with the operation of a generating unit, that is, the costs that are incurred only if the unit is generating but are independent of the amount of power generated. The first type of quasi-fixed cost is the no-load cost. If it is possible for the unit to remain connected to the system while supplying no electrical power, the no-load cost represents the cost of the fuel required to keep the unit running. Such a mode of operation is not possible for most thermal generating units. The no-load cost is then simply the constant term in the cost curve and does not have a physical meaning.

As we discussed in Chapter 2, selling at marginal cost is profitable only if this marginal cost is greater than the *average* cost of production.

4.3.1.8 Example 4.5

Let us assume that the unit of Example 4.4 is always dispatched optimally as the market price for electrical energy varies. This means that it is dispatched according to Figure 4.5. Figure 4.6 shows that its profits increase in a piecewise linear fashion with the price of electrical energy. Because of the no-load cost, the unit becomes profitable only when the price reaches 11.882 \$/MWh.

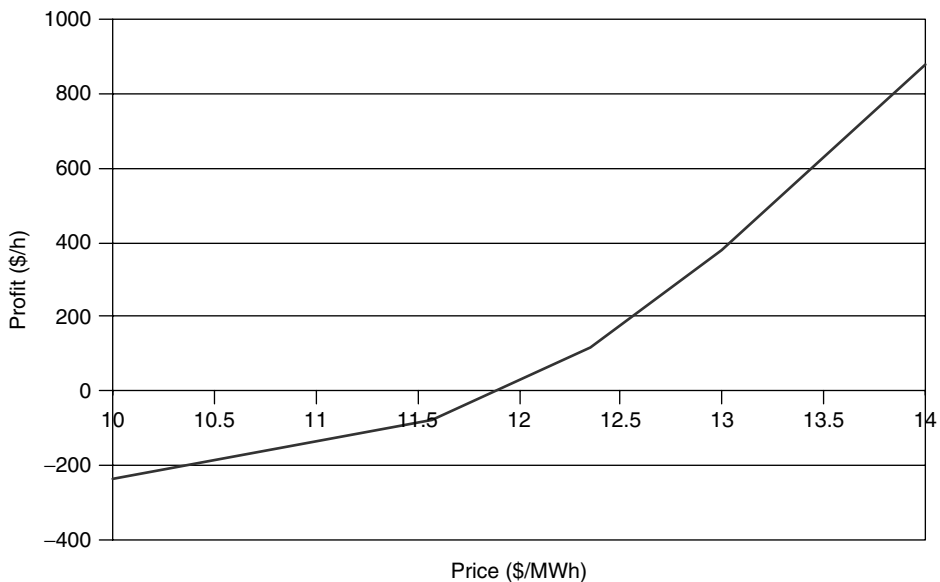


Figure 4.6 Profit accrued by the operator of the generating unit of Example 4.4 if this unit is dispatched optimally as the price of electrical energy varies

4.3.1.9 Scheduling

Since the demand for electrical energy changes over time, the price that a generator gets for its production varies. As we saw in the previous chapter, the price for electrical energy is usually constant for a period of time whose duration ranges from a few minutes to an hour depending on the market. Given a profile of prices extending over a day or more, the optimization described above could be repeated for each market period taken separately. Unfortunately, the resulting production schedule would not be optimal because it neglects the cost of starting up generating units. It would also often be technically infeasible because this approach ignores the constraints on the transitions that generating units can make between operating states. Other economic opportunities and environmental constraints may also affect the optimization of the sale of electrical energy. These different types of constraints are discussed below.

Generating units that have large start-up costs or must obey restrictive constraints will therefore not maximize their profits if their output is optimized over each period individually. Instead, their operation must be scheduled over a horizon ranging from one day to a week or more. This problem has some similarities with the unit commitment problem that monopoly utilities solve to determine how to meet a given load schedule at minimum cost with a given set of generating units. The essence of both problems is to balance the quasi-fixed and variable elements of the cost while satisfying the constraints. In the unit commitment problem, the production of all units is optimized together because their total output must be equal to the total load. On the other hand, if we assume that a generator is a price taker, its production can be optimized independently of the production of other generators. Even when this price-taking approximation holds, scheduling generation to maximize profits is computationally complex. The on/off nature of some of the decision variables makes the problem nonconvex and a rigorous treatment of the constraints significantly increases the dimensionality of the problem. Techniques such as dynamic or mixed-integer programming have been successfully used to solve this problem.

Optimally scheduling the production of a generation unit over a time horizon requires a forecast of the price of electrical energy at each period. The error inherent in such a forecast has an effect on the a posteriori optimality of the schedule. Forecasting prices accurately is extremely complex because of the number of influential factors and the lack of information on some of these factors. Since the price of electrical energy depends on the market equilibrium, it is influenced by both load and generation factors. On the load side, all the temporal, meteorological, economic and special factors that are used in load forecasting must also be taken into account when forecasting prices. The generation side is considerably more troublesome because some events occur at random (e.g. failures of generating units) and others are not always publicly announced in advance (e.g. planned outages for maintenance).

4.3.1.10 Start-up cost

The start-up cost of a generating unit represents the cost of getting this unit running and ready to produce from a shutdown state. It is thus another type of quasi-fixed cost. Diesel generators and open cycle gas turbines have low start-up costs because units of this type start quickly. On the other hand, large thermal units require a considerable

amount of heat energy before the steam is at a temperature and pressure that are sufficient to sustain the generation of electric power. These units therefore have a large start-up cost. To maximize the profitability of a thermal unit, these start-up costs must be amortized over a long period. This may even involve operating the unit at a loss for a few hours rather than shutting it down and having to re-incur the start-up cost when prices increase again.

4.3.1.11 Example 4.6

Let us examine how the coal-fired plant of Example 4.2 should be scheduled over a period of several hours. We will assume that the price at which electrical energy can be sold is set on an hourly basis and that the prices for the next few hours are shown in Figure 4.7. Suppose also that the generating unit is started up at hour 1 and that the cost of bringing it online is \$600. The table below summarizes the results.

Hour	1	2	3	4	5	6	7
Price (\$/MWh)	12.0	13.0	13.5	10.5	12.5	13.5	11.5
Generation (MW)	257.7	450.0	500.0	100.0	353.8	500.0	161.5
Revenue (\$)	3092	5850	6750	1050	4423	6750	1858
Running cost (\$)	3063	5467	6123	1235	4240	6123	1933
Start-up cost (\$)	600	0	0	0	0	0	0
Total cost (\$)	3663	5467	6123	1235	4240	6123	1933
Profit (\$)	−571	383	627	−185	183	627	−75
Cumulative profit (\$)	−571	−188	439	254	437	1064	989

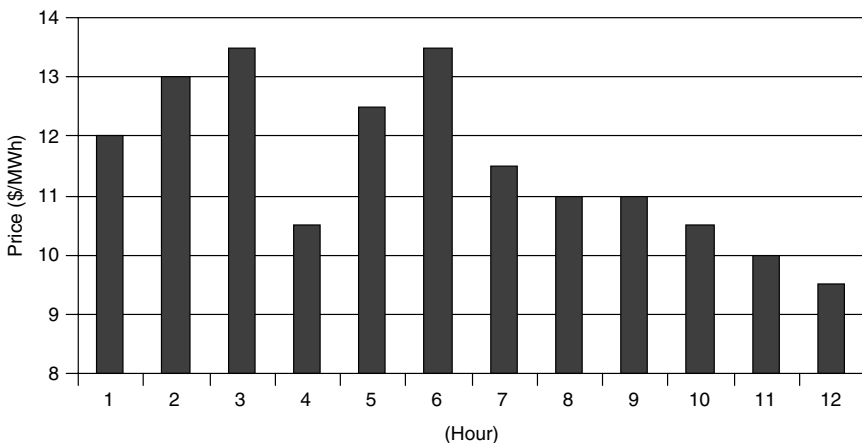


Figure 4.7 Price of electrical energy for Example 4.6

The first thing to notice is that the optimal generation varies substantially as the price of electrical energy fluctuates. The unit generates at maximum capacity at hours 3 and 6 and at minimum capacity at hour 4. The operation of this unit records a deficit at hour 1 because of the cost of starting up the unit. By hour 3, this start-up cost has been recovered and the unit begins making a profit. The price at hour 4 is so low that the unit shows a loss even though it operates at its minimum capacity. Not shutting down the unit at this hour is, however, the best decision because it avoids incurring the start-up cost again at hour 5. At hour 7, the unit records a deficit even though it is not running at its minimum capacity. This is because the unit does not generate enough to recover its no-load cost. If the price continues to decline over the next few hours, the best strategy would be to shut the unit down at the end of hour 6 and to wait until prices are higher before starting it up again.

4.3.1.12 Dynamic constraints

Starting up or shutting down a thermal generating unit or even increasing or decreasing its output by more than a small amount causes considerable mechanical stress in the prime mover. Excessive stress damages the plant and shortens its life. Limits are therefore often placed on such changes to protect these expensive assets. These safeguards have long-term benefits but short-term costs. In particular, placing a limit on the rate at which a unit can increase or decrease its output may prevent it from achieving its economically optimal output in successive periods. Minimizing the cost of these ramp-rate limits requires that the operation of the unit be optimized over at least several hours.

To limit the damage caused by frequent start-ups and shutdowns, a minimum is often placed on the number of hours that a thermal unit must remain connected to the system once it has been started. A similar limit is usually placed on the number of hours that a unit must remain idle once it has been shut down. These limits ensure that there is enough time for the temperature gradients in the turbine to subside. These minimum uptime and minimum downtime constraints reduce the opportunities to change the status of the unit and can have a profound impact on the optimal schedule. For example, the minimum downtime constraint could force a unit to continue generating at a loss during a short period of low prices because shutting it down would prevent it from reaping larger profits later on.

4.3.1.13 Environmental constraints

Generating plants must abide by environmental regulations that may affect their ability to operate at their economic optimum. Emissions of certain pollutants by fossil-fuel-fired power plants are increasingly regulated. In some cases, the rate at which a certain pollutant is released in the atmosphere is limited, thereby reducing the maximum power output of the plant. In other cases, it is the total amount of a pollutant released over a year that is capped, putting a complex integral constraint on the operation of the plant.

While hydroelectric plants do not emit pollutants and are more flexible than thermal plants, there may be constraints on their use of water. These constraints are designed to ensure the availability of water for recreation or to help the reproduction of endangered

species of fish. Water must also be made available for irrigation and for other hydroelectric plants. Optimizing the operation of hydroelectric plants is a very complex problem, particularly in river basins with multiple interconnected plants.

4.3.1.14 Other economic opportunities

The amount of electric power produced by cogeneration or combined heat and power plants is often determined by the needs of the associated industrial process. The ability of such plants to take advantage of opportunities to sell energy on the electricity market therefore may be limited.

Besides electrical energy, generators can provide other services such as reserve capacity, load following, frequency regulation and voltage regulation. These other services, which are usually called ancillary or system services, constitute a source of revenue that is distinct from the sale of electrical energy. We will discuss the issues related to the provision of these services in Chapter 5. At this point, we simply need to note that a producer's ability to trade electrical energy may be affected by contracts that it has entered into for the provision of ancillary services. Similarly, the production of electrical energy may hamper a generator's ability to provide ancillary services.

4.3.2 The production versus purchase decision

Consider the case of a generating company that has signed a contract for the supply of a given load L during a single hour. Let us first assume that this company decides to meet its contractual obligation to supply this load using its portfolio of N generating plants. It will obviously try to produce the energy required at minimum cost to itself. Mathematically, this can be formulated as the following optimization problem:

$$\text{Minimize } \sum_{i=1}^N C_i(P_i) \text{ subject to } \sum_{i=1}^N P_i = L \quad (4.9)$$

where P_i represents the production of unit i of the portfolio and $C_i(P_i)$ the cost of producing this amount of power with this unit. From calculus, we know that forming a Lagrangian function ℓ that combines the objective function and the constraint is the easiest way to solve such an optimization problem:

$$\ell(P_1, P_2, \dots, P_N, \lambda) = \sum_{i=1}^N C_i(P_i) + \lambda \left(L - \sum_{i=1}^N P_i \right) \quad (4.10)$$

where λ is a new variable called a Lagrange multiplier.

Setting the partial derivatives of this Lagrangian function to zero gives the necessary conditions for optimality and solving these equations gives the optimal solution

as follows:

$$\begin{aligned}\frac{\partial \ell}{\partial P_i} &\equiv \frac{dC_i}{dP_i} - \lambda = 0 \quad \forall i = 1, \dots, N \\ \frac{\partial \ell}{\partial \lambda} &\equiv \left(L - \sum_{i=1}^N P_i \right) = 0\end{aligned}\quad (4.11)$$

From these optimality conditions, we conclude that all the generating units in the portfolio should be operated at the same marginal cost and that this marginal cost is equal to the value of the Lagrange multiplier λ :

$$\frac{dC_1}{dP_1} = \frac{dC_2}{dP_2} = \dots = \frac{dC_N}{dP_N} = \lambda \quad (4.12)$$

The value of the Lagrange multiplier is thus equal to the cost of producing one additional megawatt-hour with any of the generating units. This Lagrange multiplier is therefore often called the *shadow price* of electrical energy.

Let us now suppose that this generating company can participate in a spot market for electricity. If the market price π is lower than the shadow price λ at which it can produce energy, our generating company should purchase energy on the market and reduce its own production up to the point at which

$$\frac{dC_1}{dP_1} = \frac{dC_2}{dP_2} = \dots = \frac{dC_N}{dP_N} = \pi \quad (4.13)$$

If the amount of energy involved is significant, the market may not have enough liquidity to handle this transaction without an increase in the price π . This issue will be discussed in more details in the next section.

4.3.2.1 Example 4.7

The 300-MW load of a small power system must be supplied at minimum cost by two thermal generating units and a small run-of-the-river hydro plant. The hydro plant generates a constant 40 MW and the cost functions of the thermal plants are given by the following expressions:

$$\text{Unit A: } C_A = 20 + 1.7 P_A + 0.04 P_A^2 \text{ \$}/\text{h}$$

$$\text{Unit B: } C_B = 16 + 1.8 P_B + 0.03 P_B^2 \text{ \$}/\text{h}$$

Since the variable operational cost of the hydro unit is negligible, the Lagrangian function of this optimization problem can be written as follows:

$$\ell = C_A(P_A) + C_B(P_B) + \lambda(L - P_A - P_B)$$

where L represents the 260 MW load that the thermal units must provide.

Setting the partial derivatives of the Lagrangian equal to zero, we obtain the necessary conditions for optimality:

$$\begin{aligned}\frac{\partial \ell}{\partial P_A} &\equiv 1.7 + 0.08 P_A - \lambda = 0 \\ \frac{\partial \ell}{\partial P_B} &\equiv 1.8 + 0.06 P_B - \lambda = 0 \\ \frac{\partial \ell}{\partial \lambda} &\equiv L - P_A - P_B = 0\end{aligned}$$

Solving this system of equations for λ , we get the marginal cost of electrical energy in this system for this loading condition:

$$\lambda = 10.67 \text{ \$/MWh}$$

We can then calculate the optimal outputs of the thermal units:

$$P_A = 112.13 \text{ MW}$$

$$P_B = 147.87 \text{ MW}$$

Replacing these values in the cost functions, we find the total cost of supplying this load:

$$C = C_A(P_A) + C_B(P_B) = 1,651.63 \text{ \$/hour}$$

4.3.3 Imperfect competition

When competition is less than perfect, some firms (the strategic players) are able to influence the market price through their actions. It is quite common for an electricity market to consist of a few strategic players and a number of price takers. A company that owns more than one generating unit is likely to have a greater influence on the market price if it optimizes the combined output of its entire portfolio of units. Optimizing the output of each unit separately would thus not maximize the firm's profit. The total profit of a firm that owns multiple generating units is thus

$$\Omega_f = \pi \cdot P_f - C_f(P_f) \quad (4.14)$$

where P_f represents the combined output of all the units controlled by that firm, while $C_f(P_f)$ represents the minimum cost at which this firm is able to produce this power. We will now assume that the market price π is no longer a variable that is beyond the control of any single market participant. Similarly, the power sold by firm f depends not only on its own decisions but also on those of its competitors. We will therefore rewrite Equation (4.14) as follows to summarize these dependencies:

$$\Omega_f = \Omega_f(X_f, X_{-f}) \quad (4.15)$$

where X_f represents the actions of firm f and X_{-f} those of its competitors.

Equation (4.15) shows that firm f cannot optimize its profits in isolation. It must consider what the other firms will do. At first sight, this may seem very difficult because these firms are competitors and exchanging information would be illegal. However, it is reasonable to make the assumption that all firms are behaving in a rational manner, that is, that they are all trying to maximize their profits. In other words, we have to find for each firm f the actions X_f^* such that

$$\Omega_f(X_f^*, X_{-f}^*) \geq \Omega_f(X_f, X_{-f}^*) \quad \forall f \quad (4.16)$$

where X_{-f}^* represents the optimal action of the other firms.

Such interacting optimization problems form what is called in game theory a *noncooperative game*. The solution of such a game, if it exists, is called a *Nash equilibrium* and represents a market equilibrium under imperfect competition.

While representing the possible actions or decisions of a firm by the generic variable X_f allowed us to formulate the problem elegantly, it hides the fact that the solution of Equation (4.16) depends on how we model the strategic interactions between the firms. In the following subsections, we discuss three models that have been proposed in the literature.

4.3.3.1 Bertrand interaction or game in prices

If we assume that the participants interact according to the Bertrand model, the price at which each firm offers its electrical energy is the only decision variable:

$$X_f = \pi_f \quad \forall f \quad (4.17)$$

The amount of energy sold by firm f is thus a function of its own offer price and the offer prices of its competitors. Firm f 's revenue is thus given by

$$\pi \cdot P_f = \pi \cdot P_f(\pi_f, \pi_{-f}^*) \quad (4.18)$$

Firm f acts as if its competitors do not change their offer prices in response to its own decisions. For an undifferentiated commodity such as electrical energy, f can sell as much as it wants as long as its price is lower than the prices of its competitors:

$$\begin{aligned} P_f(\pi_f, \pi_{-f}^*) &= P_f \text{ if } \pi_f \leq \pi_{-f}^* \\ &= 0 \quad \text{otherwise} \end{aligned} \quad (4.19)$$

As we discussed in Chapter 2, the assumption that the competition will not adjust its prices is unrealistic. Under this model, the market price will be equal to the marginal cost of production of the most efficient firm. On the one hand, no firm can offer a lower price without making a loss. On the other hand, a higher price is not sustainable because it will be undercut by the most efficient firm.

If firms can differentiate their products (e.g. if firm f sells “green” electricity while the others do not), the relation between quantity sold and price is more complex than Equation (4.19) suggests and higher prices might be sustainable.

4.3.3.2 Cournot interaction or game in quantities

In a Cournot model, each firm decides the quantity that it wants to produce

$$X_f = P_f \quad \forall f \quad (4.20)$$

The price is then determined by the inverse market demand function, which expresses the market price as a function of the total amount of energy traded

$$\pi = \pi(P_f + P_{-f}) = \pi(P) \quad (4.21)$$

If firm f assumes that its competitors will not adjust the amount of energy they produce, its revenue is given by

$$\pi \cdot P_f = \pi(P_f + P_{-f}^*) \cdot P_f \quad (4.22)$$

Its marginal revenue is therefore

$$MR_f = \frac{\partial(\pi(P) \cdot P_f)}{\partial P_f} = \pi + \frac{\partial \pi}{\partial P} \cdot P_f \quad (4.23)$$

The Cournot model suggests that firms should be able to sustain prices that are higher than the marginal cost of production, with the difference being determined by the price elasticity of the demand. Numerical results obtained with Cournot models are very sensitive to this elasticity. In particular, for a commodity like electrical energy that has a very low elasticity, the equilibrium price calculated using a Cournot model tends to be higher than the prices that are observed in the actual market.

4.3.3.3 Example 4.8

Let us consider the case of a market where two firms (A and B) compete for the supply of electrical energy. We will assume that empirical studies have shown that the inverse demand curve at a particular hour is given by

$$\pi = 100 - D \text{ \$/MWh} \quad (4.24)$$

where D is the demand for electrical energy at this hour. Let us also suppose that Firm A can produce energy more cheaply than Firm B:

$$\begin{aligned} C_A &= 35 \cdot P_A \text{ \$/h} \\ C_B &= 45 \cdot P_B \text{ \$/h} \end{aligned} \quad (4.25)$$

If we assume a Bertrand model for the competition in this market, Firm A would set its price at slightly less than the marginal cost of production of Firm B, (i.e. 45 \\$/MWh)

and would capture the whole market. At that price, the demand would be 55 MWh and Firm A would achieve a profit of \$550. At that price Firm B would lose money on any megawatt-hour that it sold and would therefore decide not to produce anything. It would then obviously not make any profit.

On the other hand, if we assume a Cournot model of competition, the state of the market is determined by the production decisions made by each firm. Let us suppose that Firms A and B have both decided to produce 5 MWh. According to the Cournot model, the market price must be such that the demand equals the total production. The total demand will be 10 MWh and, according to Equation (4.24), the market price will be 90 \$/MWh. Given the market price and the productions, we can easily find that Firms A and B make a profit of \$275 and \$225 respectively. The following cell summarizes this state of the market:

10	275
225	90

Similar cells can be generated for other combinations of productions and arranged as shown in Table 4.3. This table illustrates the interactions of the two firms under a Cournot model of competition. Toward the top left corner of the table, generators are driving the price up by limiting production. As production increases (i.e. as we move right or down through the table), the price decreases and the demand increases. Toward the bottom right corner of the table, the market is flooded and the price drops below Firm B's marginal cost of production, causing it to lose money. Among the possibilities shown in Table 4.3, Firm A would prefer the situation in which it produces 30 MWh and B produces 5 MWh because this would maximize its profit. Similarly, Firm B would like A to produce only 5 MWh so that it could produce 25 MWh and maximize its own profit. The market will not settle in either of these situations because they are not in the best interests of the other firm. Instead, the market will settle at the Nash equilibrium where neither firm can increase its profit through its own actions. The highlighted cell in Table 4.3 corresponds to this equilibrium. The profit of Firm A (\$625) is the largest that it can achieve in that row, that is, by adjusting its own production. Similarly, the profit of Firm B (\$225) is the largest in this column. Therefore, neither firm has an incentive to produce any other amount. While Firm A captures a larger share of the market because its marginal cost of production is lower, it does not freeze Firm B completely out of the market. These firms manage to maintain a price that is much higher than the marginal cost of production. This price is also higher than the value predicted by the Bertrand model.

Rather than constructing a table showing every possible pair of productions, we can formulate and solve this problem mathematically. Since each firm uses the quantity it produces as its decision variable, the profits earned by each firm are given by the following expressions:

$$\Omega_A(P_A, P_B) = \pi(D) \cdot P_A - C_A(P_A) \quad (4.26)$$

$$\Omega_B(P_A, P_B) = \pi(D) \cdot P_B - C_B(P_B) \quad (4.27)$$

where $\pi(D)$ represents the inverse demand curve. If each firm tries to maximize its profit, we have two separate optimization problems. These two optimization problems cannot be solved independently because both firms compete in the same market and the supply must be equal to the demand. Therefore we must also have

$$D = P_A + P_B \quad (4.28)$$

For each of these problems we can write a condition for optimality:

$$\frac{\partial \Omega_A}{\partial P_A} = \pi(D) - \frac{dC_A}{dP_A} + P_A \cdot \frac{d\pi}{dD} \cdot \frac{dD}{dP_A} = 0 \quad (4.29)$$

$$\frac{\partial \Omega_B}{\partial P_B} = \pi(D) - \frac{dC_B}{dP_B} + P_B \cdot \frac{d\pi}{dD} \cdot \frac{dD}{dP_B} = 0 \quad (4.30)$$

Inserting the values given by Equations (4.24) and (4.25) in Equations (4.29), (4.30), and (4.28), we get the following reaction curves:

$$P_A = \frac{1}{2}(65 - P_B) \quad (4.31)$$

$$P_B = \frac{1}{2}(55 - P_A) \quad (4.32)$$

Solving these two equations simultaneously gives the same equilibrium as what we found by building Table 4.3:

$$P_A = 25 \text{ MWh}, P_B = 15 \text{ MWh}, D = 40 \text{ MWh}, \pi = 60 \text{ \$/MWh}.$$

4.3.3.4 Example 4.9

The data from the previous example also gives us an opportunity to explore what happens when the number of firms competing in a market increases. For the sake of simplicity, we will consider the case in which Firm A competes against an increasing number of firms identical to Firm B. An optimality condition similar to Equation (4.29) or Equation (4.30) can be written for each of these firms, and this system of equations can be solved together with the inverse demand relation (4.24) and the equation expressing that all these firms compete in the same market:

$$D = P_A + P_B + \dots + P_N \quad (4.33)$$

where N represents the number of firms competing in this market. In this particular case, these equations are easy to solve for an arbitrary number of firms because firms B to N are identical and thus produce the same amount of energy. Since Firm A produces electrical energy at a lower cost than the other firms, it has a competitive advantage in this market. Figure 4.8 shows that it always produces more than any other firm does. While its share of the market decreases as the number of competing firms increases, it does not tend to zero like the individual share of the other firms. We can see from Figure 4.9 that an increase in the number of competing firms depresses the market price, even if the new firms have the same marginal cost of production as the existing

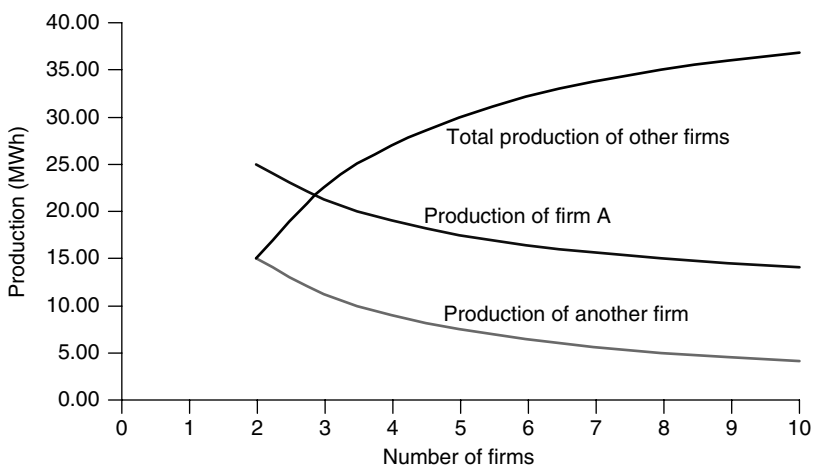


Figure 4.8 Evolution of the production of each firm as the number of competitors increases in the Cournot model of Example 4.9

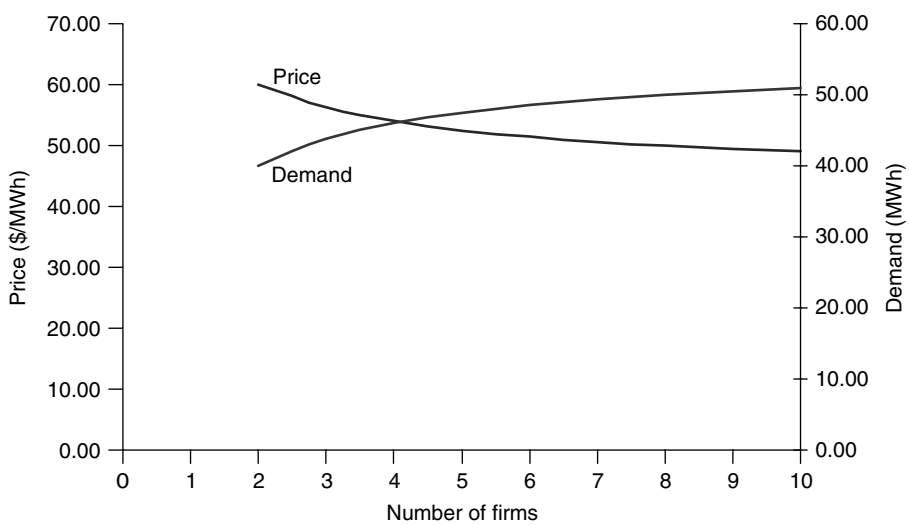


Figure 4.9 Evolution of the price and demand as the number of competitors increases in the Cournot model of Example 4.9

ones. In this case, however, the price asymptotically tends toward 40 \$/MWh, which is the marginal cost of production of firms B to N . This heightened competition induces an increase in demand and therefore benefits the consumers. Finally, as Figure 4.10 shows, this increased competition also reduces the profits made by each firm. Because of its cost advantage, the profits of Firm A are larger than the combined profits of all the other firms and, unlike the profits of the firms in the competitive fringe, it does not tend to zero as the number of competitors increases.

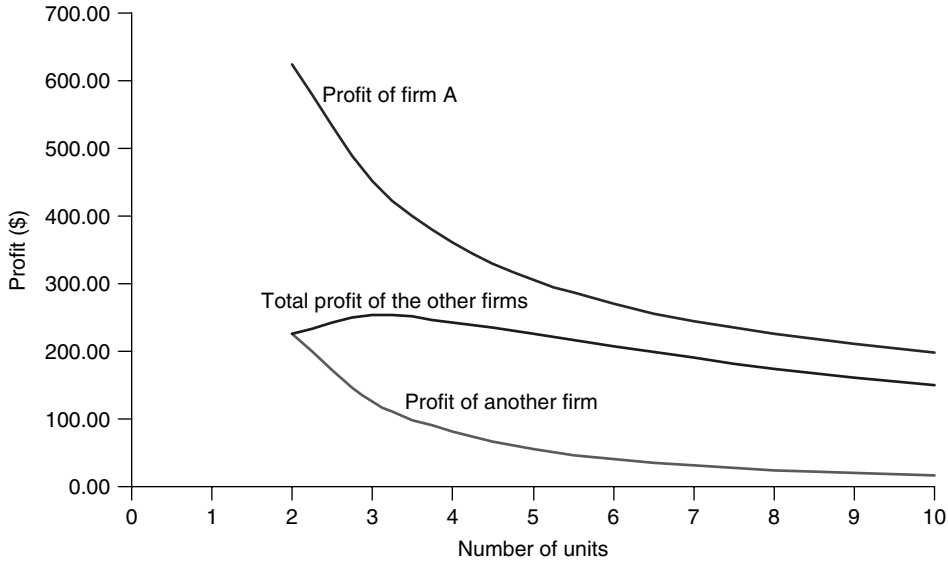


Figure 4.10 Evolution of the profits of each firm as the number of competitors increases in the Cournot model of Example 4.9

4.3.3.5 Supply functions equilibria

While the Cournot model provides interesting insights into the operation of a market with imperfect competition, its application to electricity markets produces unreasonably high forecasts for the market price. More complex representations of the strategic behavior of generating companies have therefore been developed to obtain more realistic market models. In these models, it is assumed that the amount of energy that a firm is willing to deliver is related to the market price through a *supply function*:

$$P_f = P_f(\pi) \quad \forall f \quad (4.34)$$

In this case, the decision variables of each firm are thus neither the price nor the quantity but the parameters of its supply function.

At equilibrium, the total demand is equal to the sum of the quantities produced by all the firms:

$$D(\pi) = \sum_f P_f(\pi) \quad (4.35)$$

The profit of each firm can be expressed as follows:

$$\begin{aligned} \Omega_f &= \pi \cdot P_f - C_f(P_f) \\ &= \pi \cdot \left[D(\pi) - \sum_{-f} P_{-f}(\pi) \right] - C_f \left(D(\pi) - \sum_{-f} P_{-f}(\pi) \right) \quad \forall f \end{aligned} \quad (4.36)$$

These profit functions can be differentiated with respect to the price to get the necessary conditions for optimality, which after some manipulations can be expressed in the following form:

$$P_f(\pi) = \left(\pi - \frac{dC_f(P_f)}{dP_f} \right) \cdot \left(-\frac{dD}{d\pi} + \sum_{-f} \frac{dP_{-f}(\pi)}{d\pi} \right) \quad \forall f \quad (4.37)$$

The solution of this system of equation is an equilibrium point at which all firms simultaneously maximize their profits. These optimality conditions are differential equations because the parameters of the supply functions are unknown. In order to find a unique solution to this set of differential equations, the supply and cost functions are usually assumed to have respectively linear and quadratic forms:

$$P_f(\pi) = \beta_f(\pi - \alpha_f) \quad \forall f \quad (4.38)$$

$$C_f(P_f) = \frac{1}{2}a_f P_f^2 + b_f P_f \quad \forall f \quad (4.39)$$

The decision variables are thus

$$X_f = \{\alpha_f, \beta_f\} \quad \forall f \quad (4.40)$$

The optimal values of these variables can be computed by inserting Equations (4.38) and (4.39) as well as the inverse demand function into Equation (4.37). Once these optimal values have been computed using an iterative process, it is then possible to calculate the market price, the demand, as well as the production of each firm. It is interesting to note that if the inverse demand function is affine (i.e. it includes a linear term plus an offset), the supply functions do not depend on the actual level of the demand.

4.3.3.6 Limitations of these models

Published applications to electricity markets of the models described in the previous sections have dealt so far mostly with predictions of market shares over a period of years. These models work on the aggregated capacity of each generating company and are probably not yet sophisticated enough to be useful in the daily optimization of individual generating units. In particular, they do not take into account nonlinearities such as no-load and start-up costs and dynamic constraints on the output of each unit.

Furthermore, formulating the problem as a short-run profit maximization is probably an oversimplification. In some cases, a generating company that has market power may decide to limit or even drive down the market price. Such a course of action could be justified by a desire to increase or maintain market share, by a strategy to discourage entry in the market by new participants or by a fear of attracting regulatory intervention.

4.4 Perspective of Plants with very Low Marginal Costs

Some types of plants (nuclear, hydroelectric, renewable) have negligible or almost negligible marginal costs. The challenge for the owners of such plants is to generate enough revenue to cover their very large investment costs. This causes different problems for the different types of units. Nuclear units must be operated at an almost constant generation level because adjusting their output is usually difficult. Ideally, these plants should be shut down only for refueling because of their extremely high start-up costs. The owners of nuclear power stations must therefore sell the nominal power of their units at every hour and almost at any price. On the other hand, hydroelectric plants (at least those that have a substantial reservoir) can adjust their production at will. However, the amount of energy they have available is determined by the amount of rain or snow that falls in their hydrological basin. To maximize revenues, they must therefore forecast the periods when the price paid for electrical energy will be the highest and sell energy during these periods. Finally, renewable generation depends on the availability of energy sources such as wind and sun that are not only uncontrollable but also unpredictable. Owners of such generating plants will often have to sell their production at rather unfavorable prices.

4.5 The Hybrid Participant's Perspective

A small but growing number of market participants can choose to behave like producers or consumers depending on the circumstances. Pumped hydro plants are the most common type of hybrid participant. In a traditional environment, such plants would consume energy by pumping water during periods of light load. On the other hand, during periods of high load they would produce energy by releasing this water through turbines. These cycles of consumption and production reduce the difference between the peaks and the troughs in the demand curve and hence the total cost of producing energy with thermal plants. In a competitive environment, the operation of such a plant can be profitable if the revenue generated by selling energy during periods of high prices is larger than the cost of the energy consumed during periods of low prices. This calculation must take into account the fact that, because of the losses, only about 75% of the energy consumed for pumping can be sold back.

4.5.1.1 Example 4.10

Consider a pumped hydro plant with an energy storage capacity of 1000 MWh and an efficiency of 75%. Assume that it takes four hours to completely empty or fill the upper reservoir of this plant if it operates at rated power. Suppose that the operator of this plant has decided to go through a full cycle during the 12-h period shown in Figure 4.11. A very simple strategy has been adopted: water will be pumped to the upper reservoir during the four hours with the lowest energy prices (hours 1 to 4) and will be released during the four hours with the highest energy prices (hours 7 to 10). Table 4.4 summarizes the results of this cycle. Only 750 MWh are produced

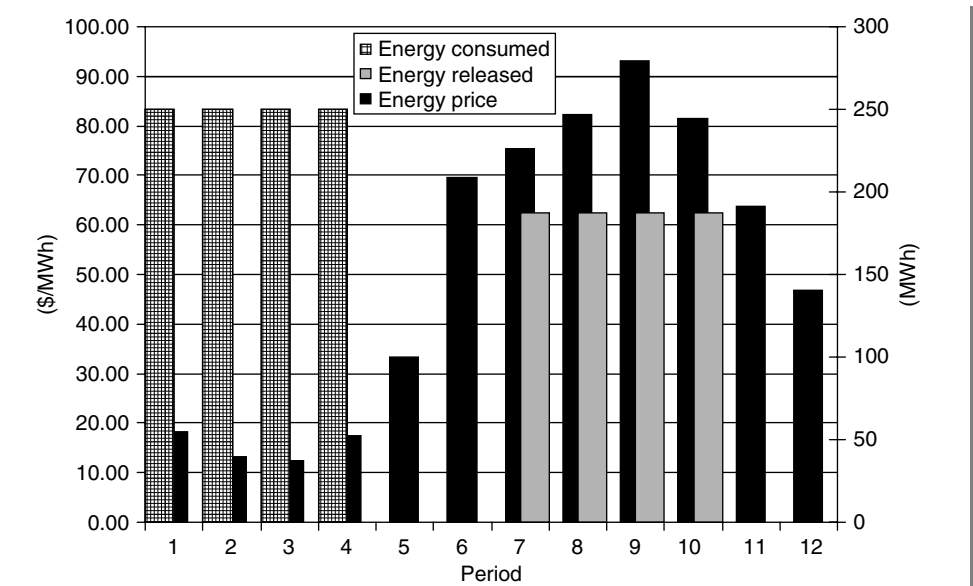


Figure 4.11 Energy prices, energy consumed and energy released for the storage plant of Example 4.10

Table 4.4 Data for Example 4.10

Period	Energy prices	Energy Consumed	Energy released	Revenue
Units	(\$/MWh)	(MWh)	(MWh)	(\$)
1	18.30	250	0	−4575
2	13.20	250	0	−3300
3	12.50	250	0	−3125
4	17.40	250	0	−4350
5	33.30	0	0	0
6	69.70	0	0	0
7	75.40	0	187.5	14 137.5
8	82.40	0	187.5	15 450
9	93.20	0	187.5	17 475
10	81.40	0	187.5	15 262.5
11	63.70	0	0	0
12	46.90	0	0	0
Total		1000	750	46 975

and resold because the plant is 75% efficient. In this case, since there are substantial differences between the periods of low prices and the periods of high prices, this cycle of pumping and turbinng results in a profit of \$46 975. If the differences in price were smaller, this profit would be substantially reduced and might even become negative.

In this example, the price of electrical energy has been taken as given. In practice, a pumped-storage plant would represent a nonnegligible portion of the load, particularly during off-peak hours. The operating strategy of the plant would therefore have to

consider the effect that it might have on prices. This type of arbitrage operations using energy storage plants is usually not profitable because the very high cost of amortizing the plant must be subtracted from the operational profits. Since pumped storage hydro plants are highly flexible, they also have the opportunity to take part in the market for ancillary services.

An increasing number of industrial consumers operate processes that cannot be shut down because of interruptions in the electricity supply without causing significant financial losses. Such consumers often install emergency generators capable of supplying at least part of their load during period of outages. When the power system operates normally, but prices are high, these consumers may find that even though the marginal cost of operating these emergency generators may be high, it is lower than the spot price of electrical energy. Under these conditions, they might want to start up their emergency generators to reduce their demand and possibly sell the surplus on the market.

Some power systems in which a competitive electricity market has been introduced are interconnected with neighboring systems that are operated by vertically integrated utilities. These utilities often take part in the competitive market. If the price paid for electrical energy is higher than their marginal cost of production, they will behave like producers in this market. On the other hand, if the price is lower than their marginal cost of production, it is in their best interest to reduce the output of their own generators and purchase power in the competitive market.

4.6 Further Reading

Bunn (2000) surveys recent developments in techniques for forecasting loads and prices, whereas Bushnell and Mansur (2001) provide some data about how consumers respond when faced with market prices for electricity. The complexity of the generation scheduling problem is explained in considerable detail (albeit in the context of a traditional utility) in (Wood and Wollenberg, 1996). Techniques for maximizing the profits of a price-taking generator are discussed in (Arroyo and Conejo, 2000 and Chan, 2000). The reader interested in supply function equilibria will find theoretical and practical discussions of this subject in (Day *et al.* (2002), Green (1996) and Klemperer (1989)). Kirschen (2003) discusses in more details demand-side participation in electricity markets.

Arroyo J M, Conejo A J, Optimal response of a thermal unit to an electricity spot market, *IEEE Transactions on Power Systems*, **15**(3), 2000, 1098–1104.

Bunn D W, Forecasting loads and prices in competitive power markets, *Proceedings of the IEEE*, **88**(2), 2000, 163–169.

Bushnell J B, Mansur E T, *The Impact of Retail Rate Deregulation on Electricity Consumption in San Diego*, Working Paper PWP-082, Program on Workable Energy Regulation, University of Californian Energy Institute, April 2001, www.ucei.org.

Chan C J S, *Development of a Profit Maximisation Unit Commitment Program*, MSc Dissertation, UMIST, 2000.

Day C J, Hobbs B F, Pang J-S, Oligopolistic competition in power networks: a conjectured supply function approach, *IEEE Transactions on Power Systems*, **17**(3), 2002, 597–607.

- Green R, Increasing competition in the British electricity spot market, *The Journal of Industrial Economics*, **XLIV**(2), 1996, 205–216.
- Kirschen D S, A demand-side view of electricity markets, *IEEE Transactions on Power Systems*, **18**(2), 2003, 520–527.
- Klemperer P D, Meyer M A, Supply function equilibria in oligopoly under uncertainty, *Econometrica*, **57**(6), 1989, 1243–1277.
- Wood A J, Wollenberg B F, *Power Generation, Operation and Control*, Second Edition, John Wiley & Sons, New York, 1996.

4.7 Problems

- 4.1 Cheapo Electrons is an electricity retailer. The table below shows the load that it forecast its consumers would use over a 6-h period. Cheapo Electrons purchased in the forward market and the power exchange exactly enough energy to cover this forecast. The table shows the average price that it paid for this energy for each hour. As one might expect, the actual consumption of its customers did not exactly match the load forecast and it had to purchase or sell the difference on the spot market at the prices indicated. Assuming that Cheapo Electrons sells energy to its customers at a flat rate of 24.00 \$/MWh, calculate the profit or loss that it made during this 6-h period. What would be the rate that it should have charged its customers to break even?

Period	1	2	3	4	5	6
Load Forecast (MWh)	120	230	310	240	135	110
Average cost (\$/MWh)	22.5	24.5	29.3	25.2	23.1	21.9
Actual load (MWh)	110	225	330	250	125	105
Spot price (\$/MWh)	21.6	25.1	32	25.9	22.5	21.5

- 4.2 The input–output curve of a gas-fired generating unit is approximated by the following function:

$$H(P) = 120 + 9.3 P + 0.0025 P^2 \text{ MJ/h}$$

This unit has a minimum stable generation of 200 MW and a maximum output of 500 MW. The cost of gas is 1.20 \$/MJ. Over a 6-h period, the output of this unit is sold in a market for electrical energy at the prices shown in the table below.

Period	1	2	3	4	5	6
Price (\$/MWh)	12.5	10	13	13.5	15	11

Assuming that this unit is optimally dispatched, is initially on-line and cannot be shut down, calculate its operational profit or loss for this period.

- 4.3 Repeat the calculation of Problem 4.2 assuming that the cost curve is replaced by a three-segment piecewise linear approximation whose values correspond

with those given by the quadratic function for 200 MW, 300 MW, 400 MW and 500 MW.

- 4.4 Assume that the unit of Problem 4.2 has a start-up cost of \$500 and that it is initially shut down. Given the same prices as in Problem 4.2, when should this unit be brought on-line and when should it be shut down to maximize its operational profit? Assume that dynamic constraints do not affect the optimal dispatch of this generating unit.
- 4.5 Repeat Problem 4.4, taking into account that the minimum uptime of this unit is four hours.
- 4.6 Borduria Generation owns three generating units that have the following cost functions:

$$\text{Unit A: } 15 + 1.4 P_A + 0.04 P_A^2 \text{ \$/h}$$

$$\text{Unit B: } 25 + 1.6 P_B + 0.05 P_B^2 \text{ \$/h}$$

$$\text{Unit C: } 20 + 1.8 P_C + 0.02 P_C^2 \text{ \$/h}$$

How should these units be dispatched if Borduria Generation must supply a load of 350 MW at minimum cost?

- 4.7 How would the dispatch of Problem 4.6 change if Borduria Generation had the opportunity to buy some of this energy on the spot market at a price of 8.20 \$/MWh?
- 4.8 If, in addition to supplying a 350-MW load, Borduria Generation had the opportunity to sell energy on the electricity market at a price of 10.20 \$/MWh, what is the optimal amount of power that it should sell? What profit would it derive from this sale?
- 4.9 Repeat Problem 4.8 if the outputs of the generating units are limited as follows:

$$P_A^{\text{MAX}} = 100 \text{ MW}$$

$$P_B^{\text{MAX}} = 80 \text{ MW}$$

$$P_C^{\text{MAX}} = 250 \text{ MW}$$

- 4.10 Consider a market for electrical energy that is supplied by two generating companies whose cost functions are

$$C_A = 36 \cdot P_A \text{ \$/h}$$

$$C_B = 31 \cdot P_B \text{ \$/h}$$

The inverse demand curve for this market is estimated to be

$$\pi = 120 - D \text{ \$/MWh}$$

Assuming a Cournot model of competition, use a table similar to the one used in Example 4.8 to calculate the equilibrium point of this market (price, quantity, production and profit of each firm).

(Hint: Use a spreadsheet. A resolution of 5 MW is acceptable)

- 4.11 Write and solve the optimality conditions for Problem 4.10.
- 4.12 Consider the pumped hydro plant of Example 4.10 and the price profile shown in the table below. Assuming that the operator uses the same strategy as in the example (reservoir initially empty, pumping during four hours of lowest prices and turbinning during four hours of highest prices), calculate the profit or loss that this plant would make during this cycle of operation. Determine the value of the plant efficiency that would make the profit or loss equal to zero.

Period	1	2	3	4	5	6
Price (\$/MWh)	40.92	39.39	39.18	40.65	45.42	56.34
Period	7	8	9	10	11	12
Price (\$/MWh)	58.05	60.15	63.39	59.85	54.54	49.50